

- PCB Stackups
- LAYER 1 : TOP
 - LAYER 2 : GND
 - LAYER 3 : IN1
 - LAYER 4 : IN2
 - LAYER 5 : PWR
 - LAYER 6 : BOT

15.6" Discrete Block Diagram

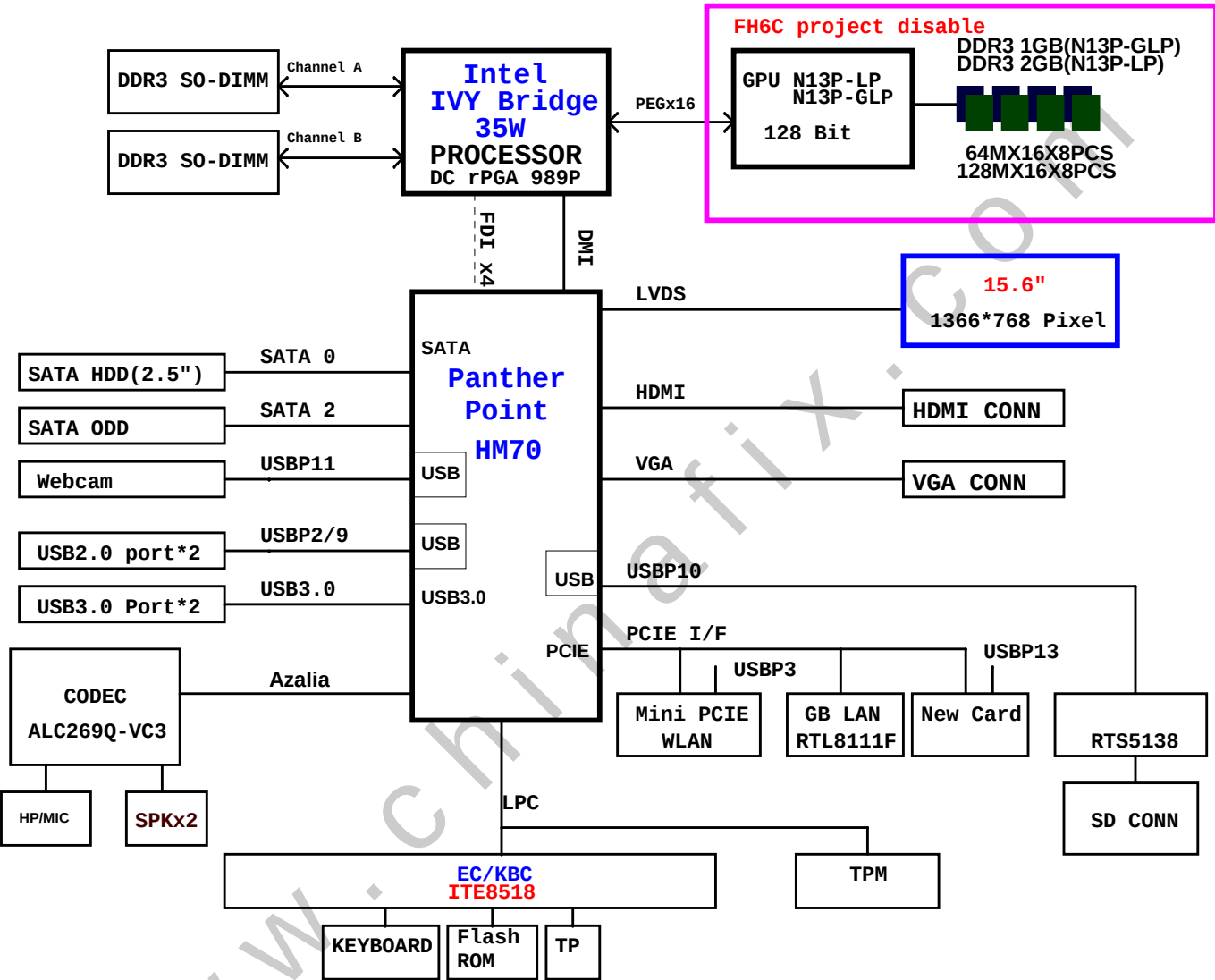
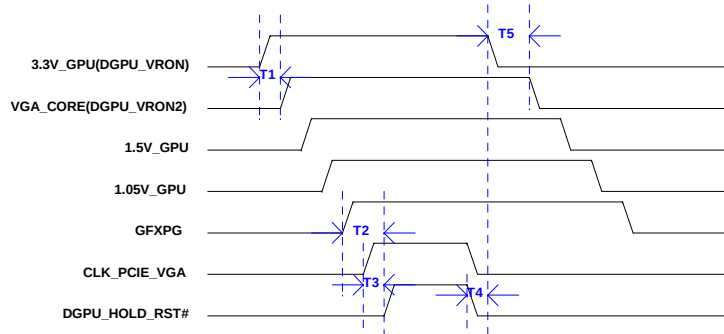


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POWER PLANE	VOLTAGE	CONTROL SIGNAL	Power States ACTIVE IN
VIN	10V~+19V		S0-S5
3V_RTC	+3.0V~+3.3V		S0-G3
3V_S0	+3.3V	S0_ON1	S0
3V_S5	+3.3V	EC	S0-S5
3V_AUX	+3.3V	AC/DC Insert enable	AWLAYS
5V_S0	+5V	S0_ON1	S0
5V_S3	+5V	S3_ON	S0-S3
5V_S5	+5V	EC	S0-S5
5V_AUX	+5V	AC/DC Insert enable	AWLAYS
1.8V_S0	+1.8V	S0_ON2	S0
1.5V_S0	+1.5V	S0_ON2	S0
1.5V_S3	+1.5V	S3_ON	S0-S3
1.05V_S0	+1.05V	S0_ON2	S0
VCCSA	By VID	S0_ON2	S0
CPU_CORE	By VID	VR_ON	S0
VCC_AXG	By VID	VR_ON	S0
3V_LAN	+3.3V	LAN_ON	S0-S5(By WOL)
3V_GPU	+3.3V	DGPU_VRON	Optimus
1.5V_GPU	+1.5V	DGFX_VR_PWRGD	Optimus
1.05V_GPU	+1.05V	DGFX_VR_PWRGD	Optimus
VGA_CORE	By VID	DGPU_VRON1	Optimus

N13P-LP Power ON/OFF Sequence



BIOS/ EC control:

T1:DGPU_VRON to DGPU_VRON2 = 500us

T2:GFXPG to DGPU_HOLD_RST# = 5ms

T3:CLK_PCIE_VGA to DGPU_HOLD_RST# >100us(Spec)

T4:DGPU_HOLD_RST# to DGPU_VRON = 5ms

Note: Clock must be shutdown before 3.3V_GPU

T5:DGPU_VRON to DGPU_VRON2 = 500us

N13P-LP & N13P-GLP Table

	N13P-GLP	N13P-LP
VL3	BLM18P121SN (CX8PG121009)	0ohm_0603 (CS00003J951)

	N13P-GLP	N13P-LP
VR111	NA	10Kohm_0402 (CS31002FB26)

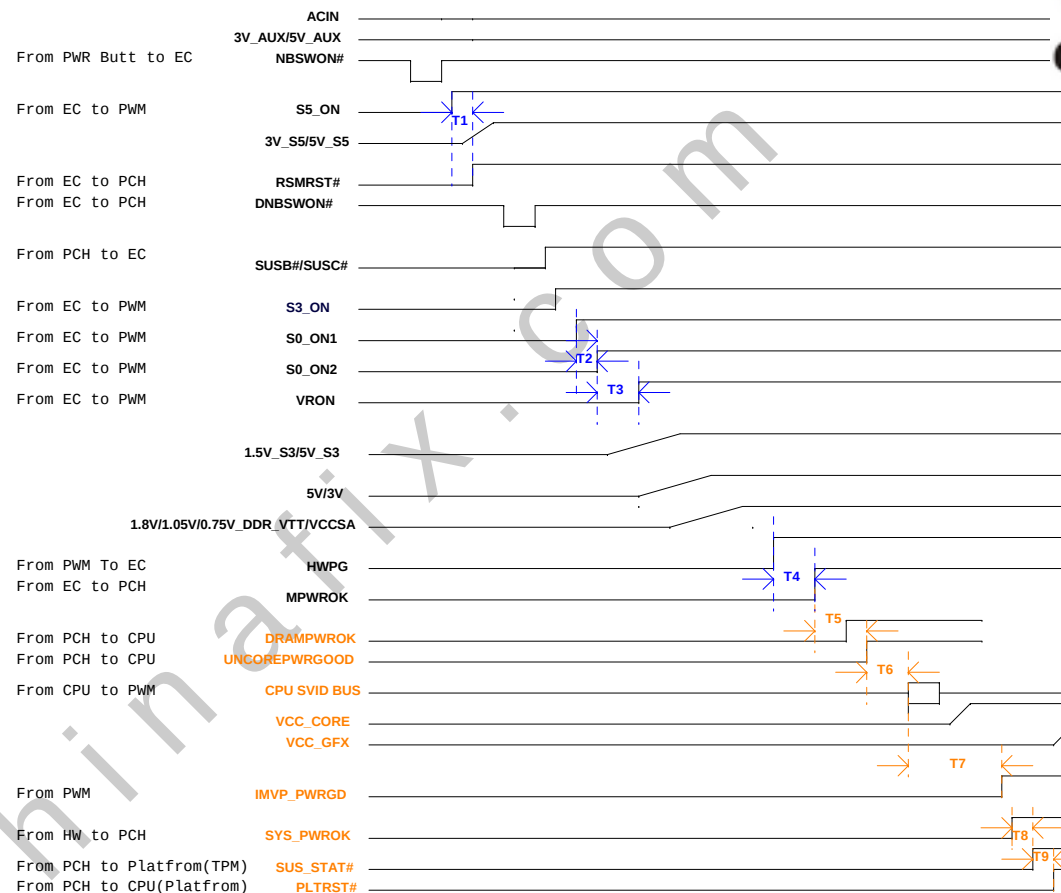
	N13P-GLP	N13P-LP
VR62	10Kohm_0402 (CS31002FB26)	NA

ID2	ID1	ID0	Model PU/PD RES P/N:CS31002JB28
0-R435	0-R438	0-R437	FJ8 UMA
0-R435	0-R438	1-R430	FJ8 Discrete
0-R435	1-R382	0-R437	FH6 UMA(Consumer)
0-R435	1-R382	1-R430	FH6 UMA(Commercial)
1-R384	0-R438	0-R437	FH6 N13P-LP
1-R384	0-R438	1-R430	FH6 N13P-GLP
1-R384	1-R382	0-R437	TBD
1-R384	1-R382	1-R430	TBD

B-29

		GLP 1GB HYN	GLP 1GB SAM	GLP 2GB HYN	GLP 2GB SAM	LP 2GB HYN	LP 2GB SAM
ROM_SCLK	VR44 VR54	NA CS31502FB24	NA CS31502FB24	NA CS31502FB24	NA CS31502FB24	CS24992FB26 NA	CS24992FB26 NA
ROM_SI	VR41 VR52	NA CS31502FB24	NA CS32002FB29	NA CS33012FB18	NA CS34532FB18	NA CS33012FB18	NA CS34532FB18
ROM_SO	VR43 VR53	NA CS31002FB26	NA CS31002FB26	NA CS31002FB26	NA CS31002FB26	CS31002FB26 NA	CS34532FB18 NA
STRAP0	VR51 VR55	CS34532FB18 NA	CS34532FB18 NA	CS34532FB18 NA	CS34532FB18 NA	CS34532FB18 NA	CS34532FB18 NA
STRAP1	VR46 VR56	NA CS34532FB18	NA CS34532FB18	NA CS34532FB18	NA CS34532FB18	NA CS24992FB26	NA CS24992FB26
STRAP2	VR47 VR57	CS24992FB26 NA	CS24992FB26 NA	CS24992FB26 NA	CS24992FB26 NA	CS24992FB26 NA	CS32002FB29 NA
STRAP3	VR48 VR58	NA CS24992FB26	NA CS24992FB26	NA CS24992FB26	NA CS24992FB26	NA CS24992FB26	NA CS24992FB26
STRAP4	VR50 VR59	NA NA	NA NA	NA NA	NA NA	NA CS34532FB18	NA CS34532FB18

System Power-ON Sequence



System Power Sequence

EC Control:

T1: S5_ON TO RSMRST# = 20ms (spec:mini 10ms)

T2: S0_ON1 TO S0_ON2 = 500us

T3: S0_ON2 TO VRON = 10ms

T4: HWP TO MPWROK = 110ms (spec:mini 99ms)

Note:HWP NEED TO BE HIGH at that time

System:

T5: MPWROK to UNCOREPWROK =2ms(Min)

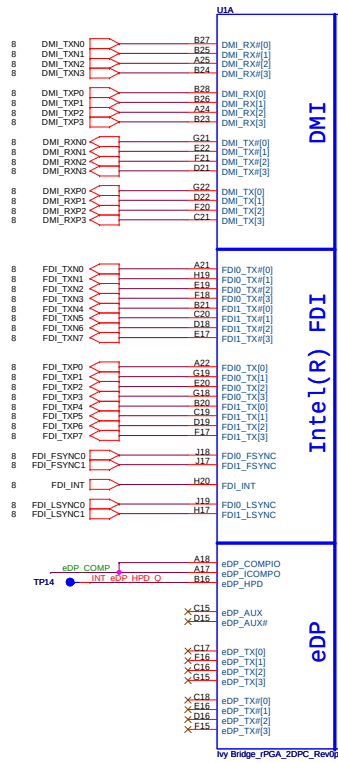
T6: UNCOREPWROK to SVID Packet =500us(Max)

T7: SVID Packet to IMVP_PWRGD =5ms(Max)

T8: SYS_PWROK to SUS_STAT# =1ms(Min)

T9:SUS_STAT# to PLTRST# =60us(Min)

Ivy Bridge Processor (DMI,PEG,FDI)



Intel(R) FDI

PCI EXPRESS * - GRAPHICS

Intel(R) FDI

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Intel(R) FDI

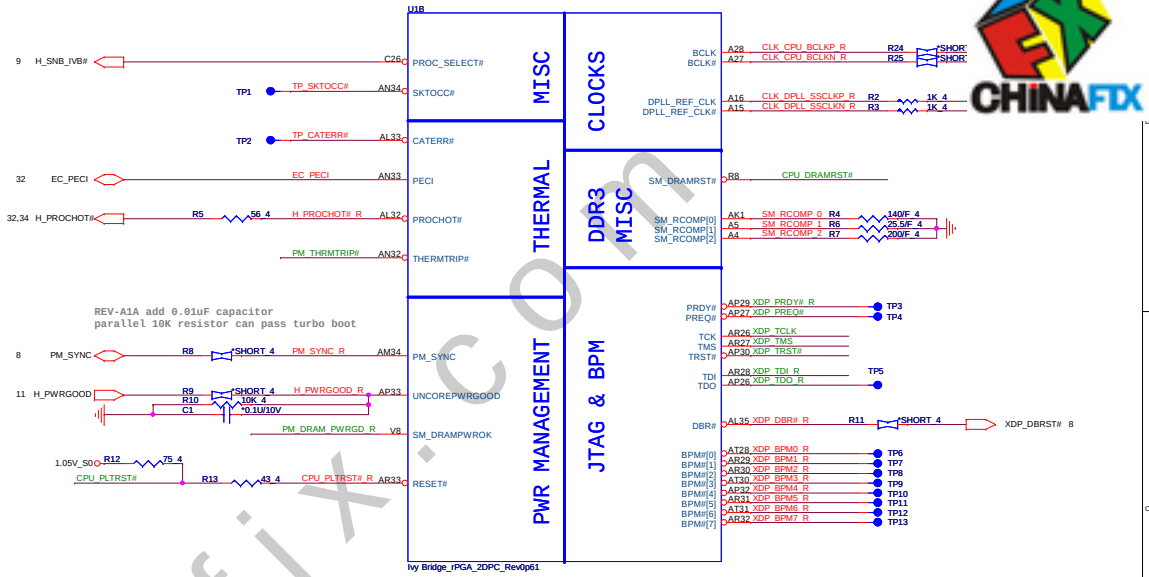
PCI EXPRESS * - GRAPHICS

Intel(R) FDI

PCI EXPRESS * - GRAPHICS

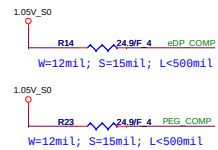
Ivy Bridge_PGA_ZDPC_Rev0p61

Ivy Bridge Processor (CLK,MISC,JTAG)

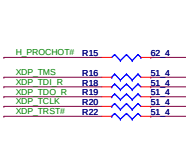


Ivy Bridge_PGA_ZDPC_Rev0p61

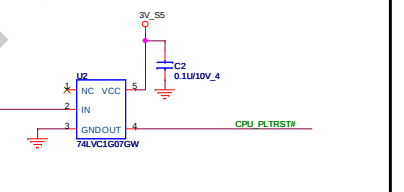
DP & PEG Compensation



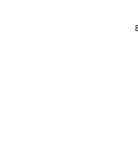
Processor pull-up



Level Shift

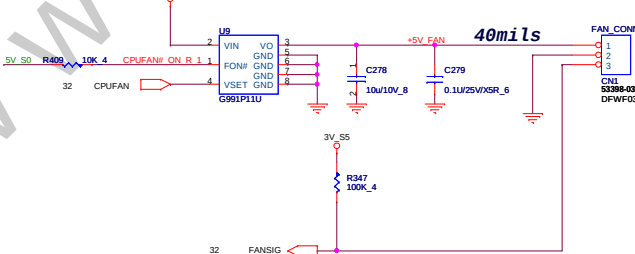


Thermal Trip

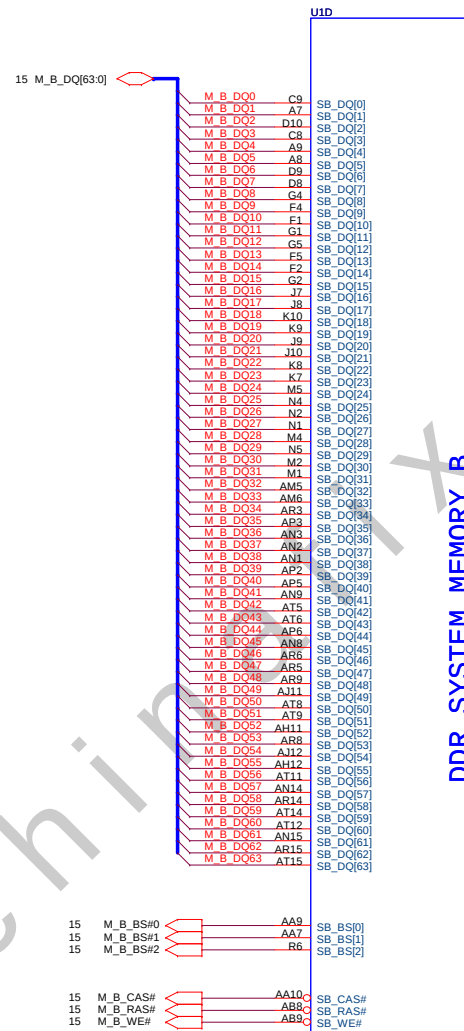


FAN Control-->For one FAN solution

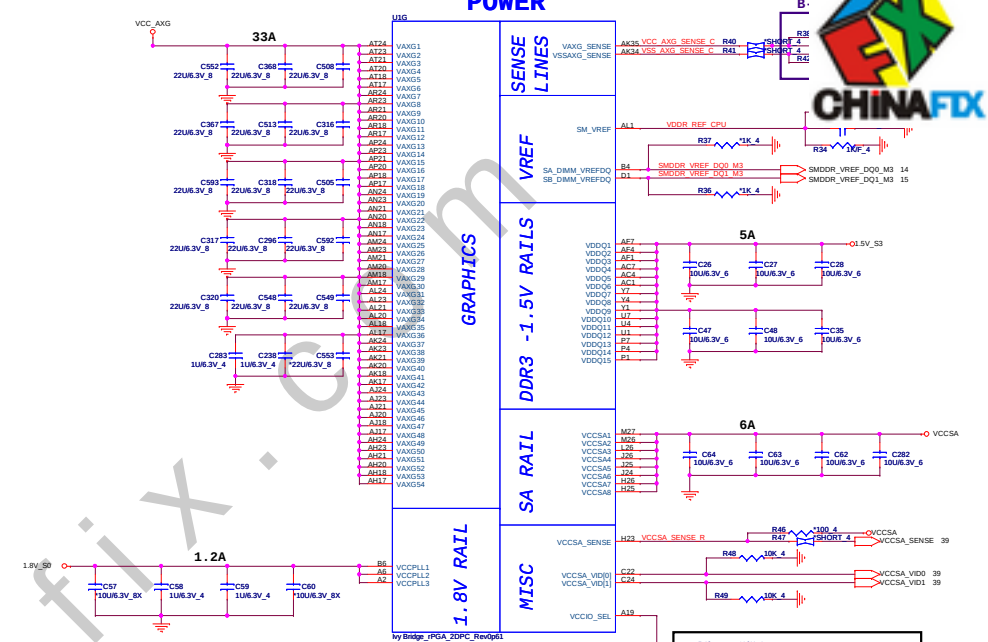
VSET >= 1V, Enable
FANPWR = 1.6*VSET



05



Ivy Bridge Processor (GRAPHIC POWER)

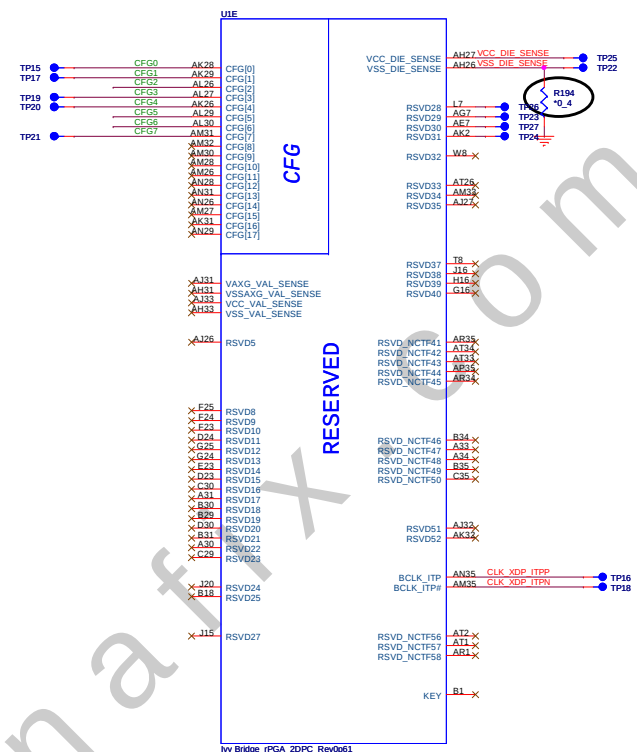


The diagram shows a 3V_S0 supply connected to a resistor network. The network consists of a 4kΩ resistor in series with a parallel combination of a 10kΩ resistor and a 4kΩ resistor. The output voltage is labeled VCCIO_SEL.

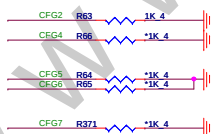
VCCIO_SEL	
1	1.05V
0	0V

The schematic diagram illustrates the VR_SVID interface connections. It consists of three signal lines: VR_SVID_CLK (34), VR_SVID_DATA (34), and VR_SVID_ALERTF (34). Each line is connected to a CPU pin (H_CPU_SVIDCLK, H_CPU_SVIDDAT, H_CPU_SVIDALRTF) through a resistor (R58, R60, R62) and a 100kΩ pull-up resistor (R59, R61, R63) to a 1.05V supply. The CPU pins are also connected to a 100kΩ pull-down resistor (R57, R59, R61) to ground.

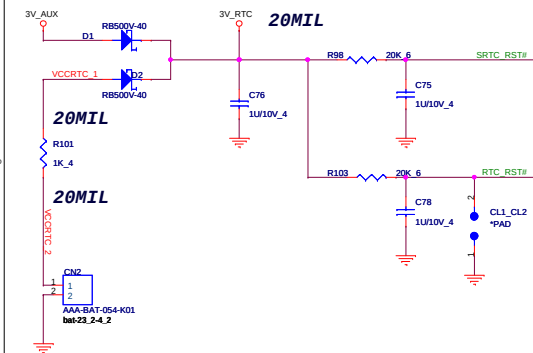
Ivy Bridge Processor (RESERVED, CFG)



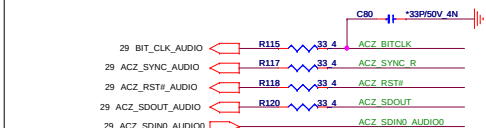
CFG#			Setting
CFG2	0	PCIe X16 LANE Reversed	
	1	Normal Operation	
CFG3	0	PCIe X4 LANE Reversed	
	1	Normal Operation	
CFG4	0	Enable; An ext DP device is connected to eDP	
	1	Disable; No physical DP attached to eDP	
CFG(5:6)	00	1 x 8 , 2 x 4 PCIe	
	01	Reserve	
	10	2 x 8 PCIe	
	11	1 x 16 PCIe	
CFG7	0	PEG Wait for BIOS for training	
	1	PEG Train immediately following PLT_RST#	



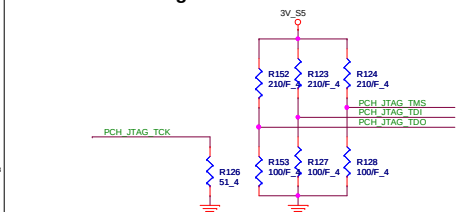
RTC Circuit



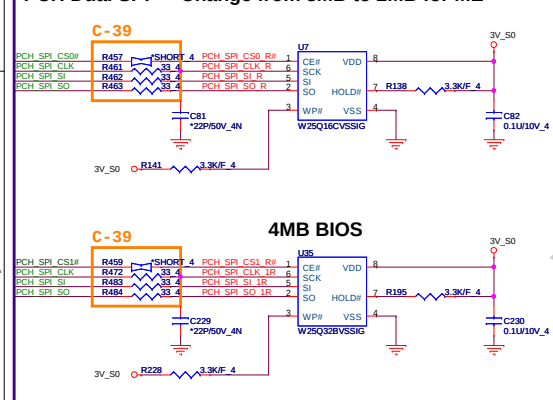
HDA Bus



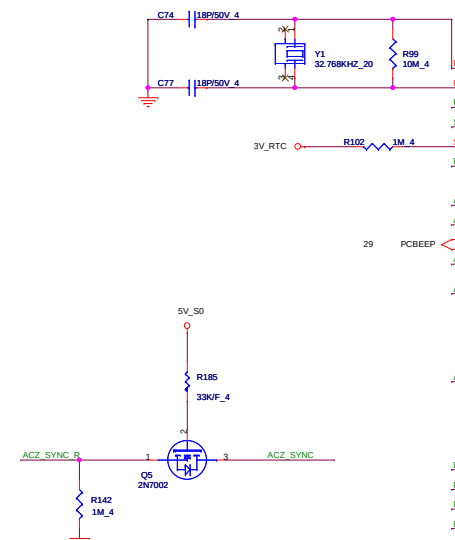
PCH JTAG Debug



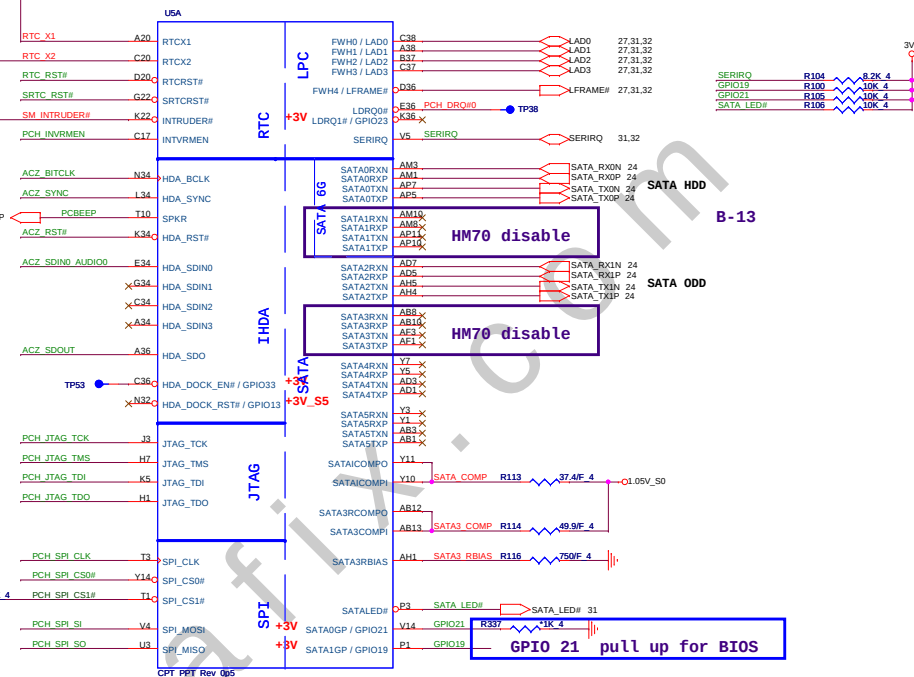
PCH Dual SPI	Change from 8MB to 2MB for ME
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PCH2



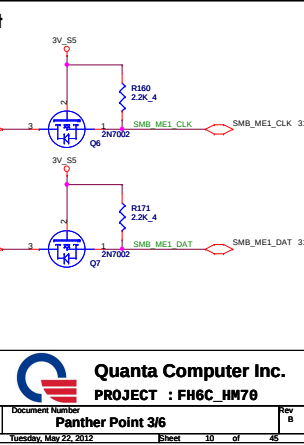
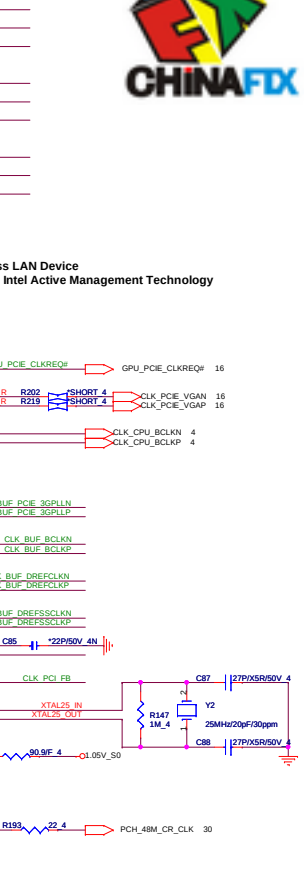
Panther Point (HDA, JTAG, SATA)



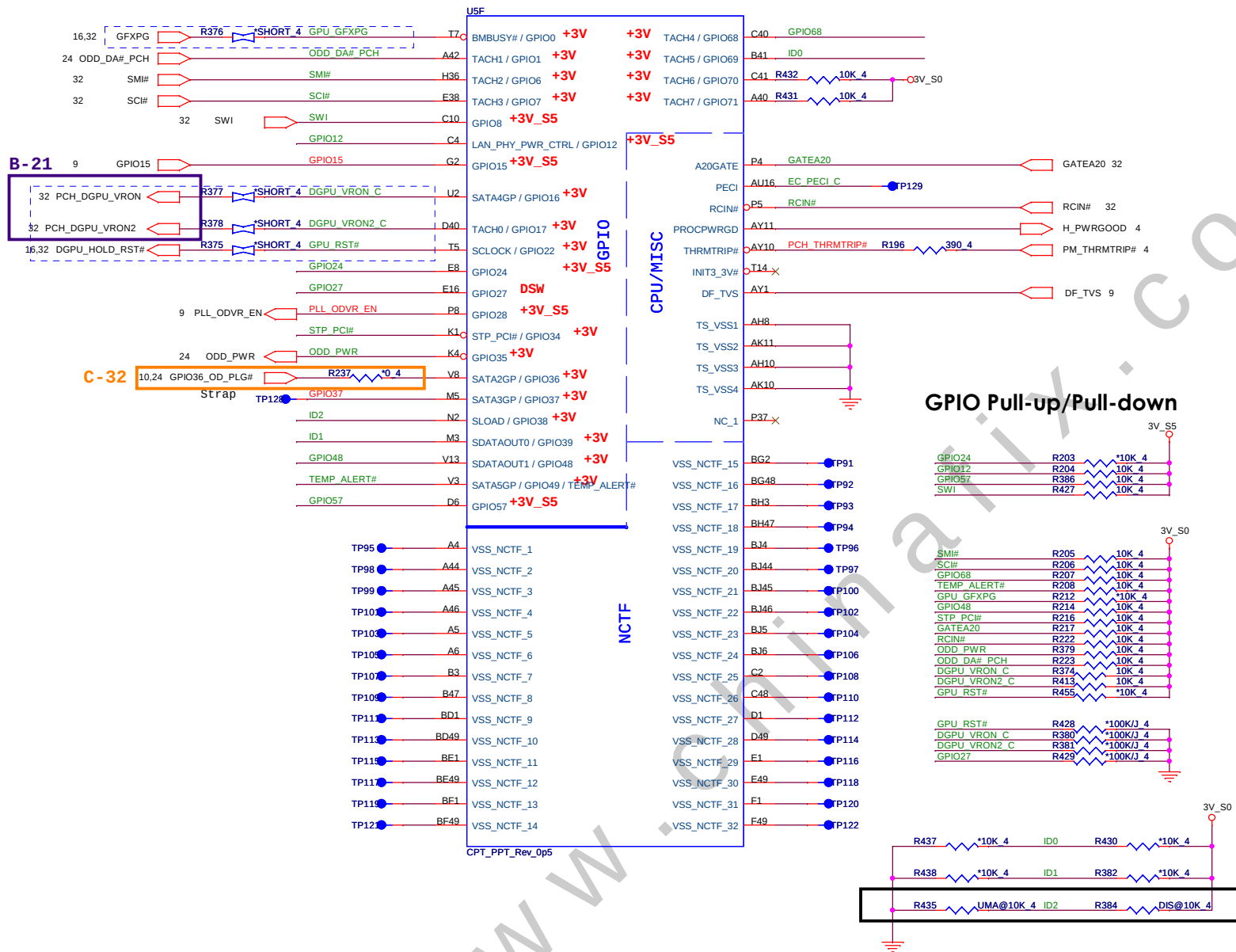
PCH Strap Table

Pin Name	Strap description	Sampled	Configuration										
SPKR	No reboot mode setting	PWROK	0 = Default (weak pull-down 20K) 1 = Setting to No-Reboot mode	3V_S0 PCBEEP									
GNT3# / GPIO55	Top-Block Swap Override	PWROK	0 = "top-block swap" mode 1 = Default (weak pull-up 20K)	PCI_GNT3# 10									
INTVRMEN	Integrated 1.05V VRM enable	ALWAYS	Should be always pull-up	3V_RTC PCH_INVRMEN									
GNT1# / GPIO51	Boot BIOS Selection 1 [bit-1]	PWROK	<table border="1"><thead><tr><th>GNT1#</th><th>GPIO19</th><th>Boot Location</th></tr></thead><tbody><tr><td>1</td><td>1</td><td>SPI</td></tr><tr><td>0</td><td>0</td><td>LPC</td></tr></tbody></table>	GNT1#	GPIO19	Boot Location	1	1	SPI	0	0	LPC	GNT1# 10
GNT1#	GPIO19	Boot Location											
1	1	SPI											
0	0	LPC											
GPIO19	Boot BIOS Selection 0 [bit-0]	PWROK		GPIO19									
HDA_SDO	Flash Descriptor Security	RSMRST	1 = Override 0 = Default (weak PD 20K)	3V_S0 ACZ_SDOOUT ACZ_SDOOUT 32									
DF_TV5	DMI/FDI Termination voltage	PWROK	0 = Set to Vss 1 = Set to Vcc (weak pull-down 20K)	OL_BV_S0 DF_TV5 11									
GPIO28	On-die PLL Voltage Regulator	RSMRST#	0 = Disable 1 = Enable (Default)	3V_AUX0 PLL_ODVR_EN 11									
HDA_SYNC	On-Die PLL VR Voltage Select	RSMRST	0 = Support by 1.8V (weak pull-down) 1 = Support by 1.5V	3V_S5 ACZ_SYNC									
GPIO15	TLS Confidentiality	RSMRST	0 = Default, TLS no Confidentiality 1 = TLS Confidentiality	3V_S5 GPIO15 11									
DSWVRMEN	Deep S4/S5 Well On -Die Voltage Regulator Enable	ALWAYS	0 = Disable 1 = Enable	3V_RTC DSWVRMEN 8									
INIT3_3V#	Reserved	PWROK	1 = Default (weak pull-up 20K)	Should not pull low. leave as No Connect									
GNT2# / GPIO53	ESI Strap (Server Only)	PWROK	1 = Default, Should not be pulled low for desktop and mobile	Should not pull low for desktop and mobile									
L_DDC_DATA	LVDS Detected	PWROK	0 = Default, Not Detected 1 = Detected	1= PU to 3V									
SDVO_CTRLDATA	Port B Detected	PWROK	0 = Default, Not Detected 1 = Detected	1= PU to 3V									
DDPC_CTRLDATA	Port C Detected	PWROK	0 = Default, Not Detected 1 = Detected	0=NC									
DDPD_CTRLDATA	Port D Detected	PWROK	0 = Default, Not Detected 1 = Detected	0=NC									
SATA3GP / GPIO37	Reserved	PWROK	0 = Default	Should not be pulled high when strap is sampled									
SATA2GP / GPIO36	Reserved	PWROK	0 = Default	Should not be pulled high when strap is sampled									

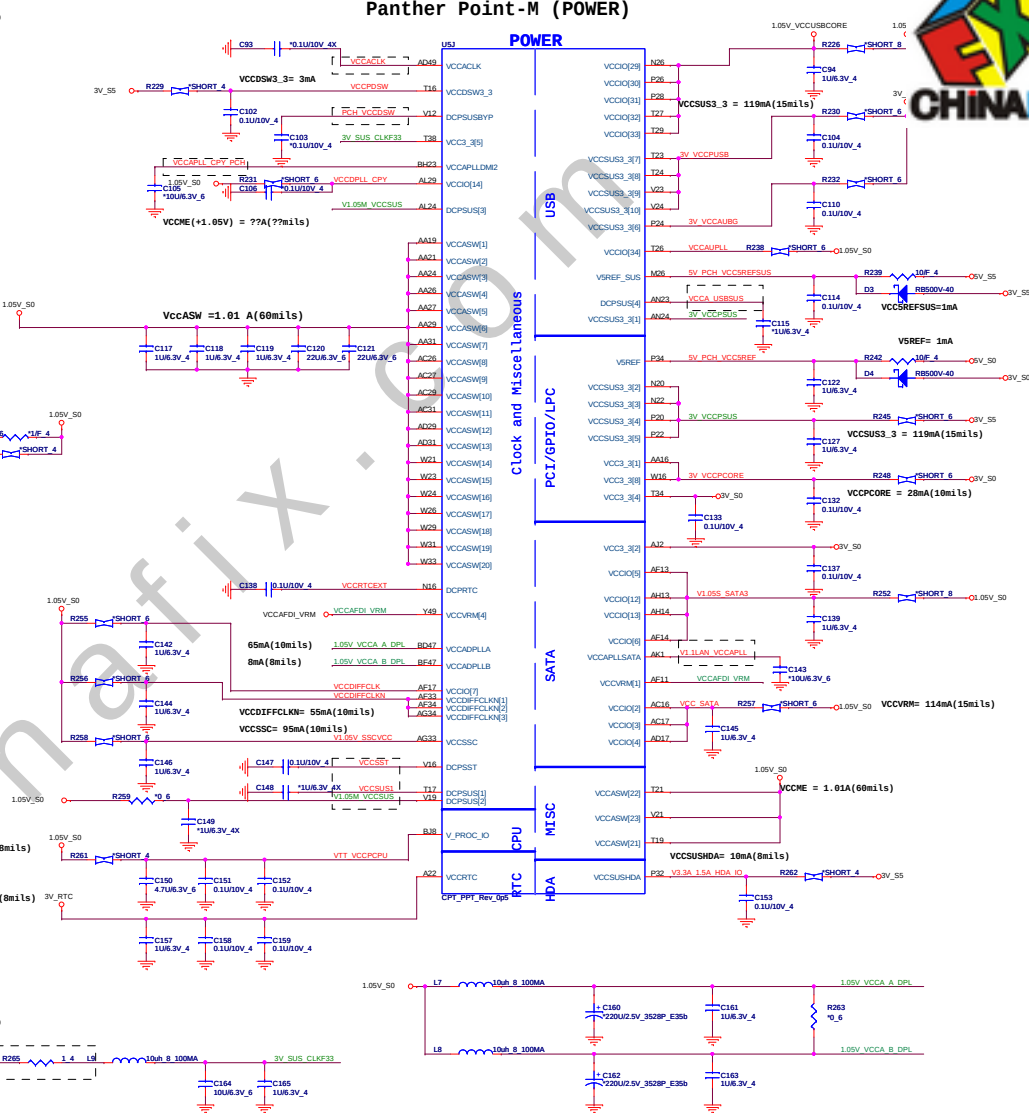
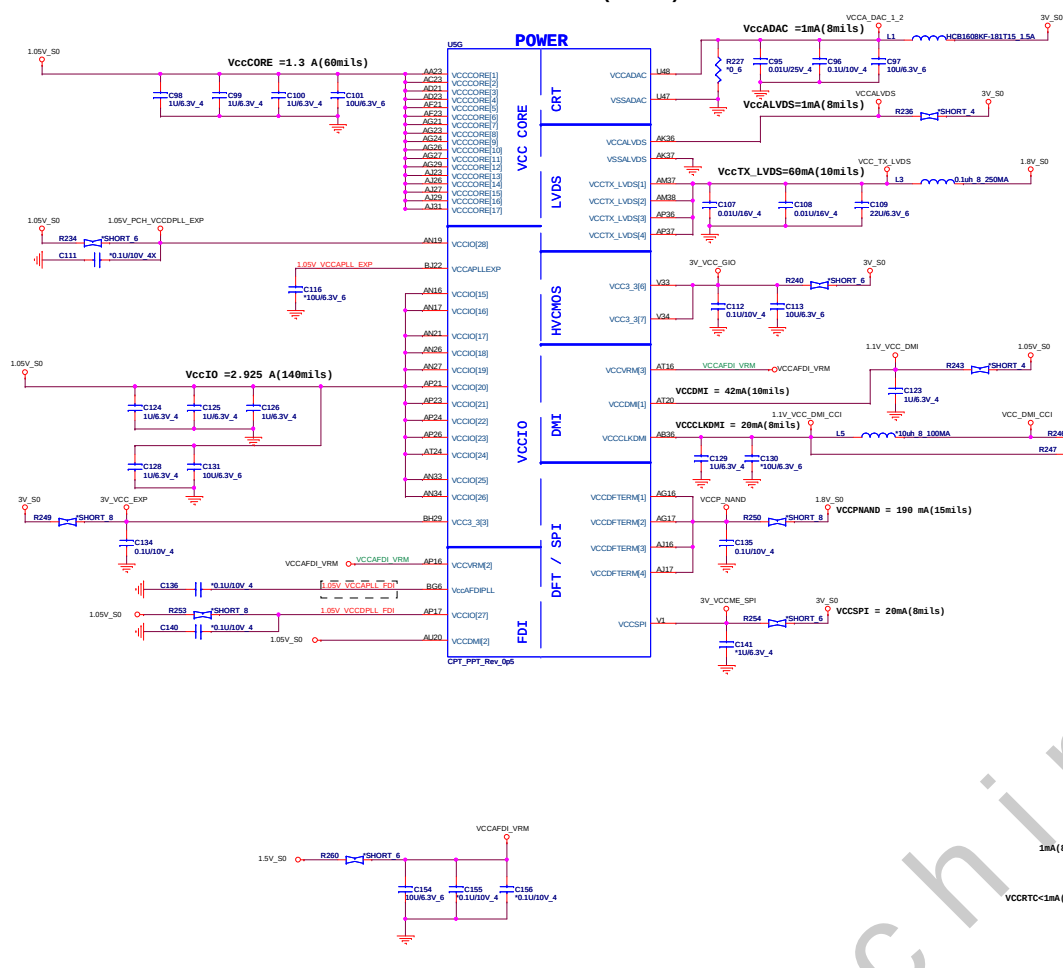
Panther Point-M (PCI-E, SMBUS, CLK)



Panther Point (GPIO, VSS_NCTF, RSVD)



ID2	ID1	ID0	Model
0	0	0	FJ8 UMA
0	0	1	FJ8 Discrete
0	1	0	FH6 UMA(Consumer)
0	1	1	FH6 UMA(Commercial)
1	0	0	FH6 N13P-LP
1	0	1	FH6 N13P-GLP
1	1	0	TBD
1	1	1	TBD

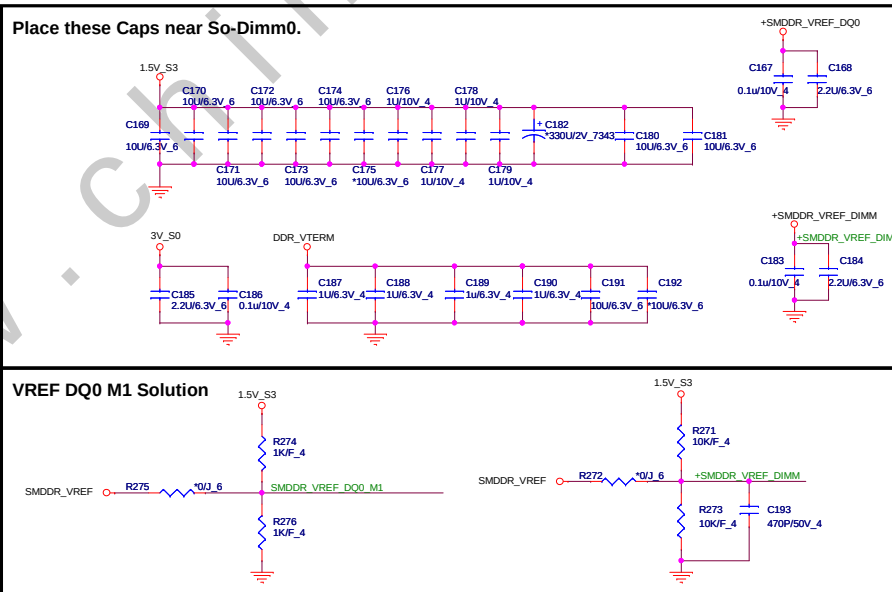
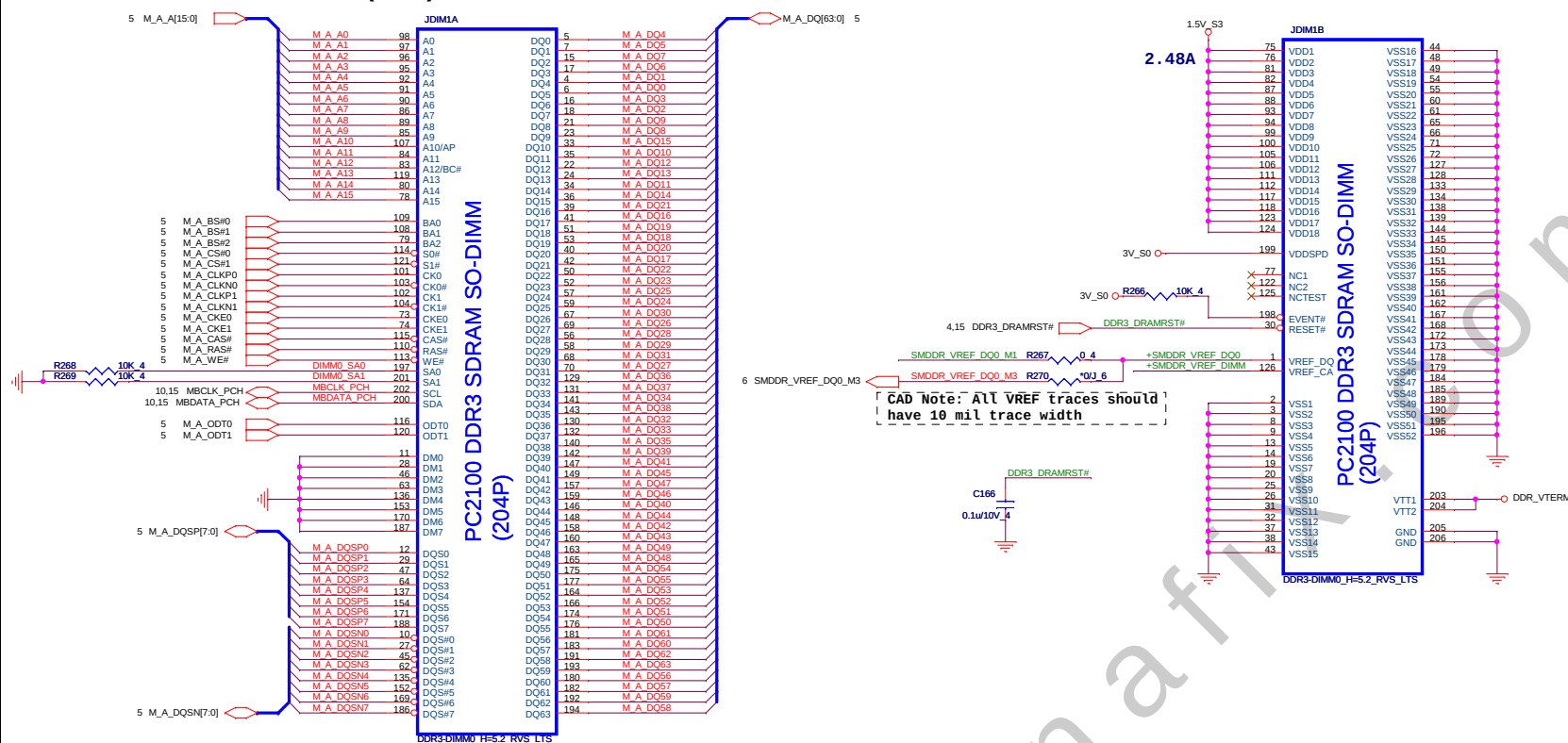




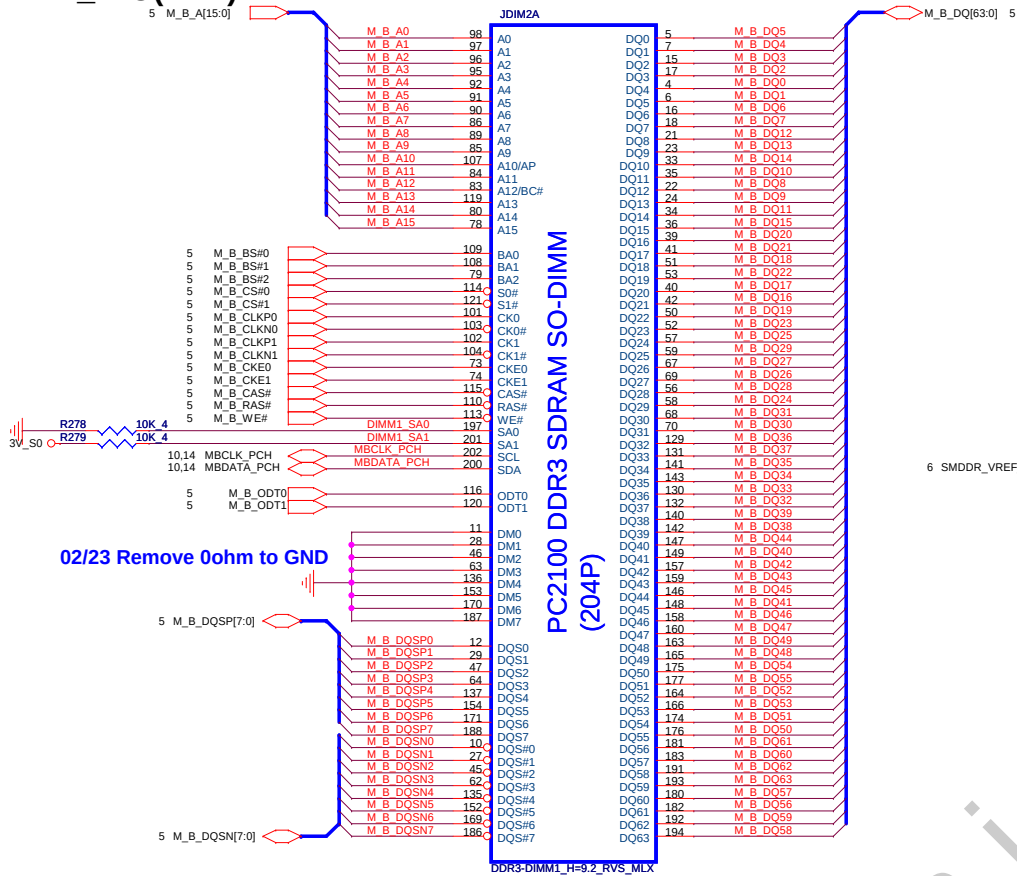
Quanta Computer Inc.
PROJECT : FH6C_HM70



DDR_STD(DDR)



DDR_RVS(DDR)

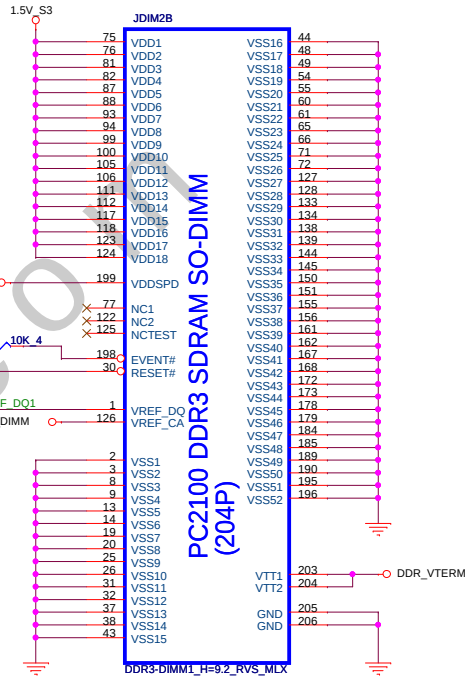
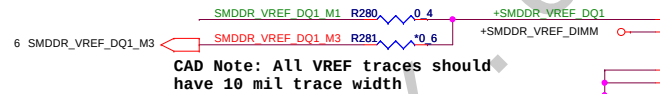


02/23 Remove 0ohm to GND

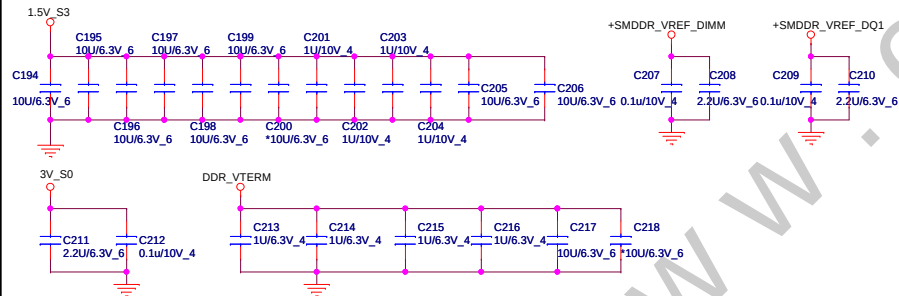
PC2100 DDR3 SDRAM SO-DIMM (204P)

DDR3-DIMM1_H=9.2_RVS_MLX

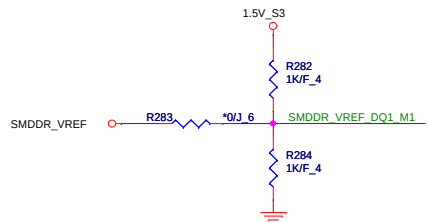
2.48A



Place these Caps near So-Dimm1.

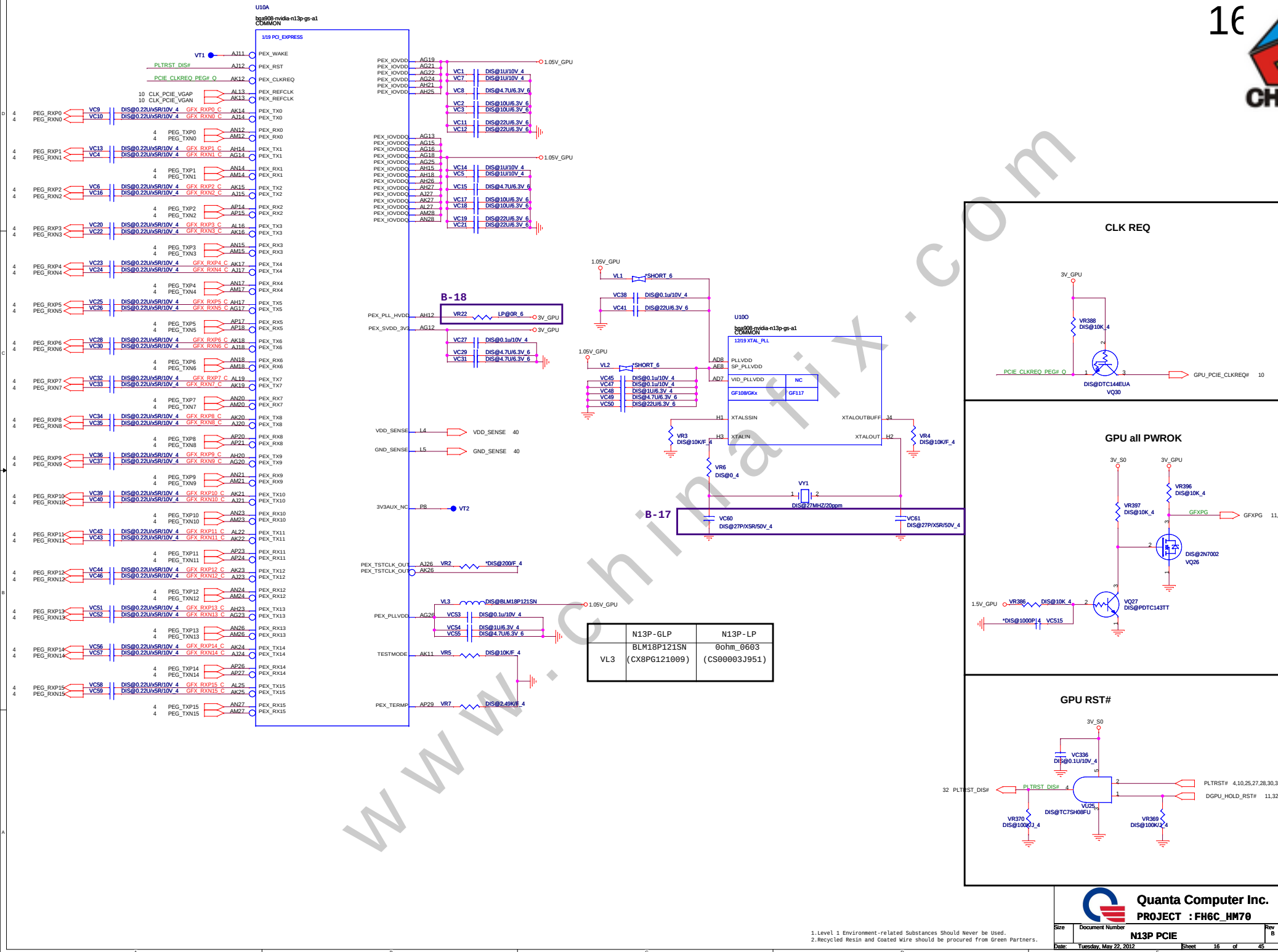


VREF DQ1 M1 Solution



Quanta Computer Inc.
PROJECT : FH6C_HM70

Size	Document Number	Rev
	DDRIII SO-DIMM-1	B
Date:	Tuesday, May 22, 2012	Sheet 15 of 45



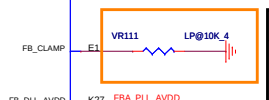
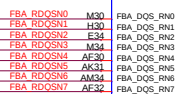
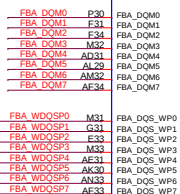


U10B

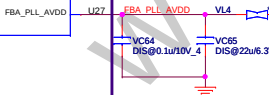
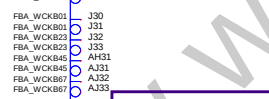
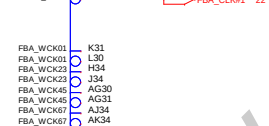
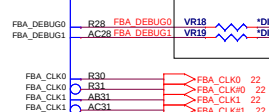
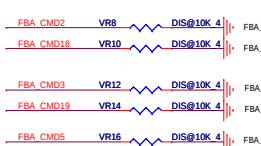
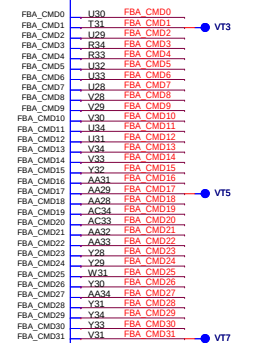
bga908-nvidia-n13p-gs-a1

COMMON

219 FBA



	N13P-GLP	N13P-LP
VR111	NA	10Kohm_0402 (CS31002FB26)



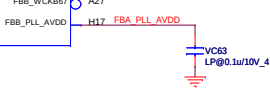
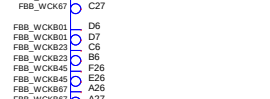
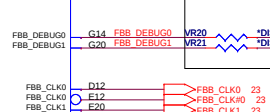
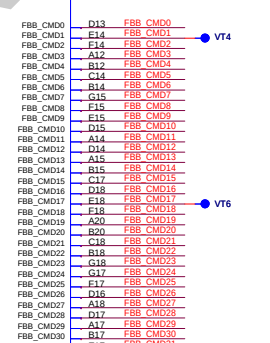
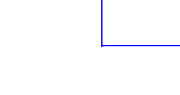
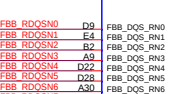
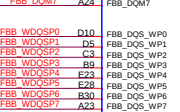
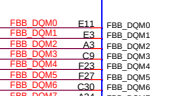
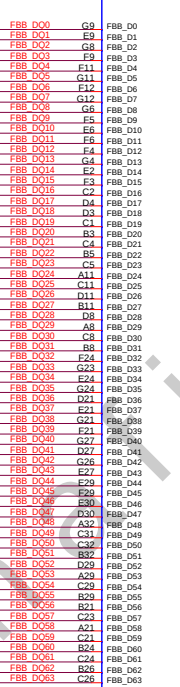
B-07

U10C

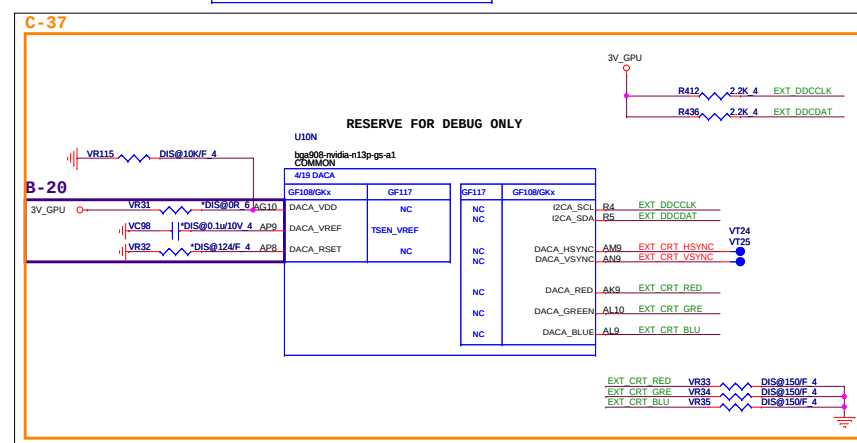
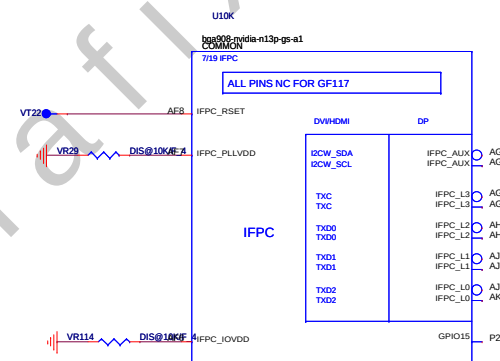
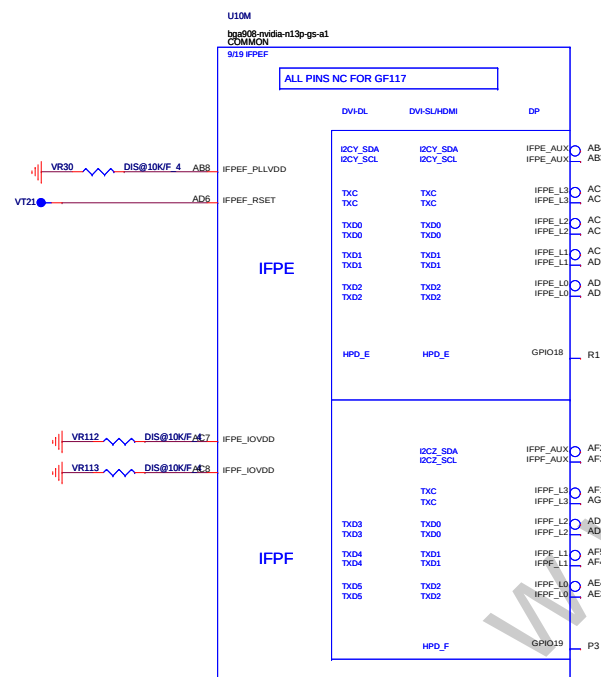
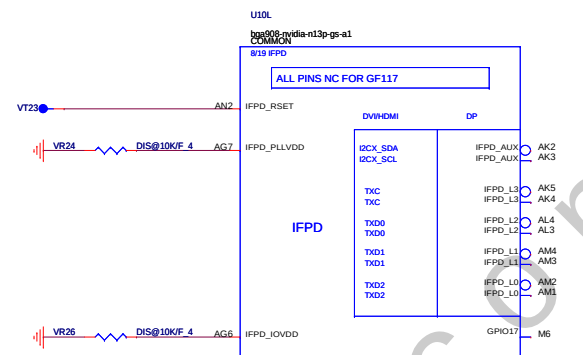
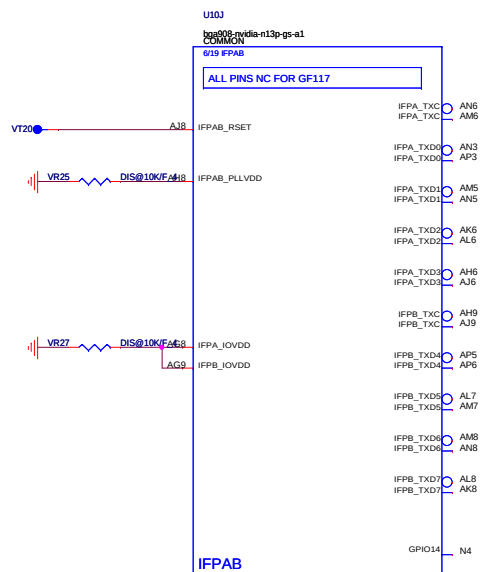
bga908-nvidia-n13p-gs-a1

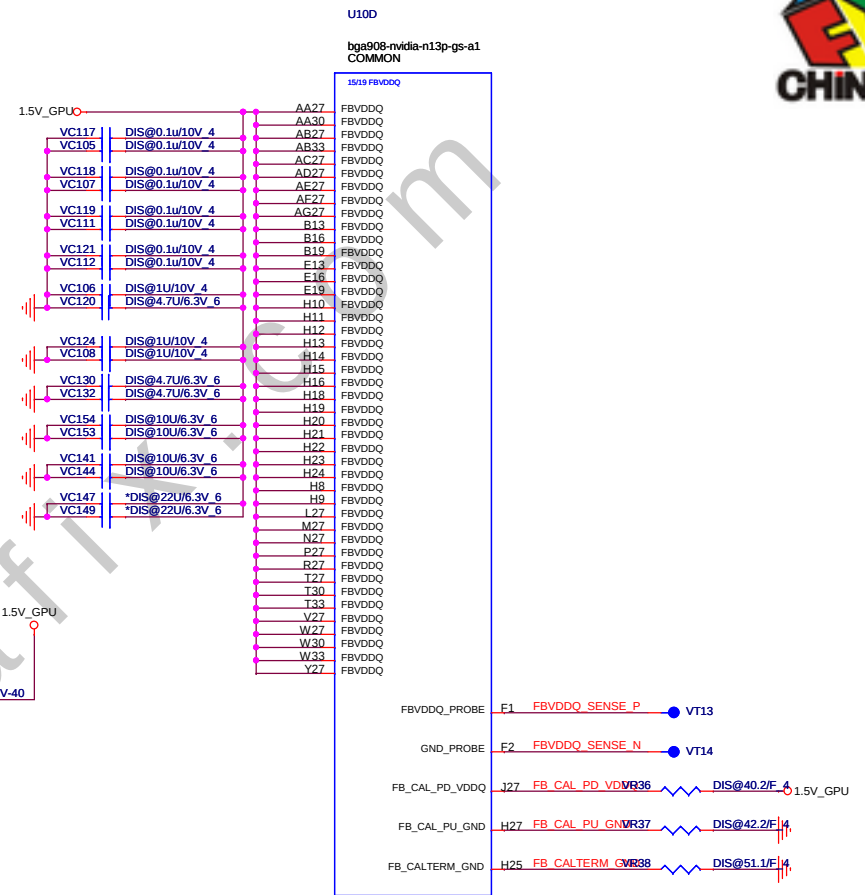
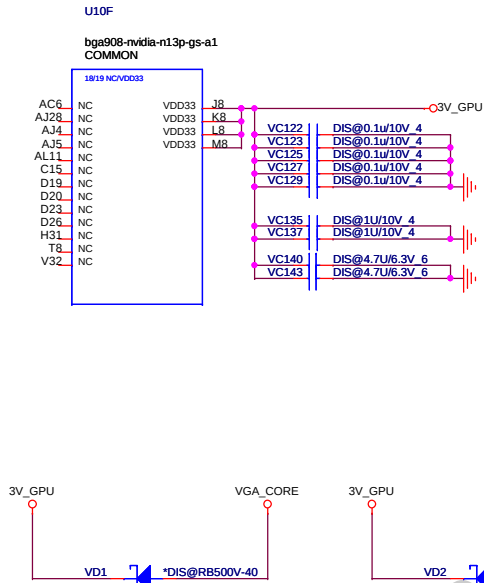
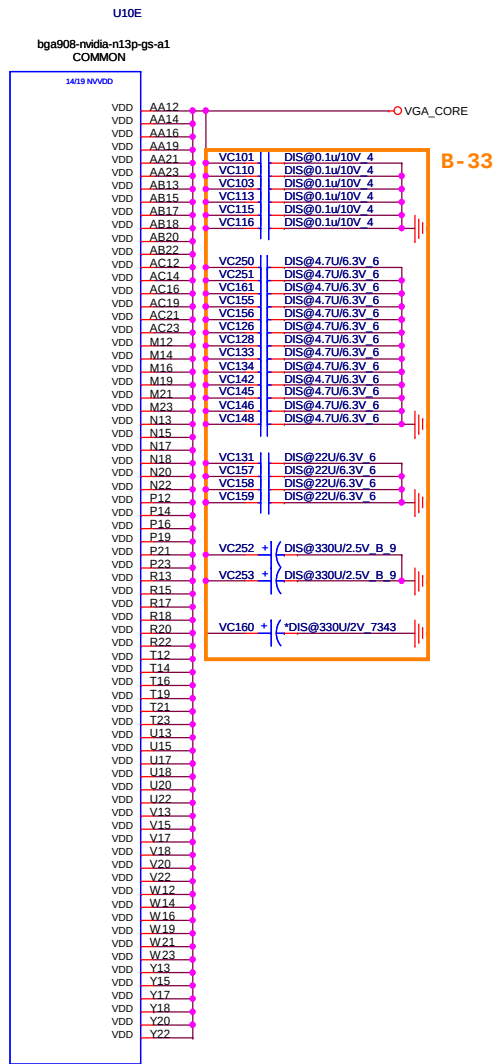
COMMON

319 FBB



Quanta Computer Inc.
PROJECT : FH6C_HM70

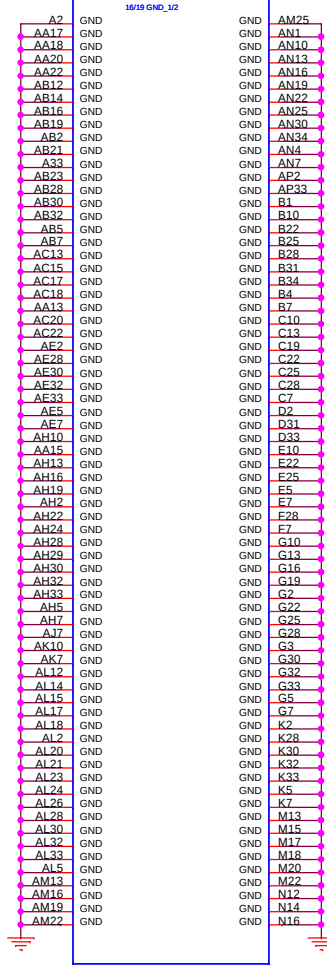






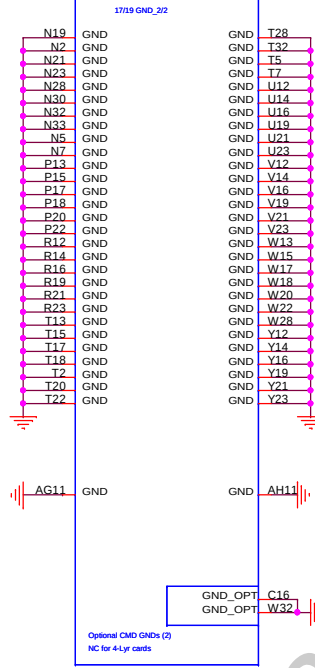
U10G

bga908-nvidia-n13p-gs-a1
COMMON



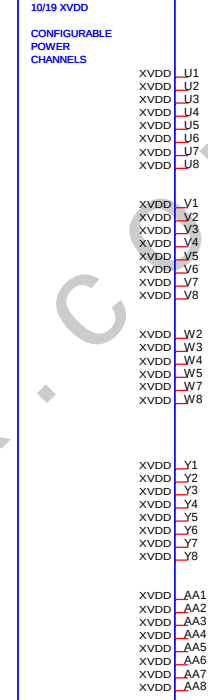
U10I

bga908-nvidia-n13p-gs-a1
COMMON



U10H

bga908-nvidia-n13p-gs-a1
COMMON



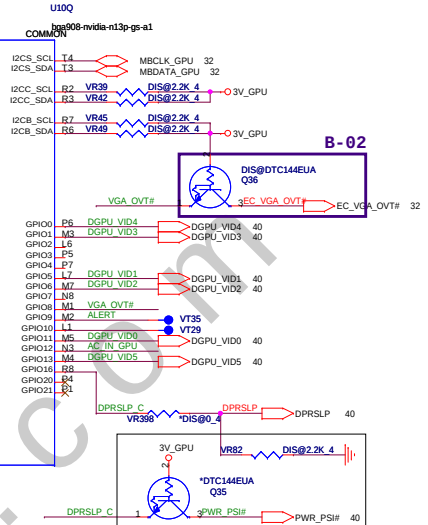
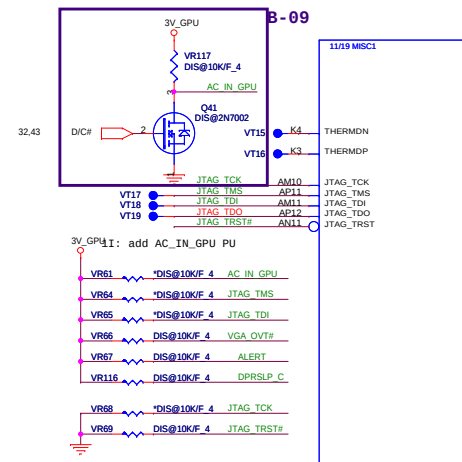
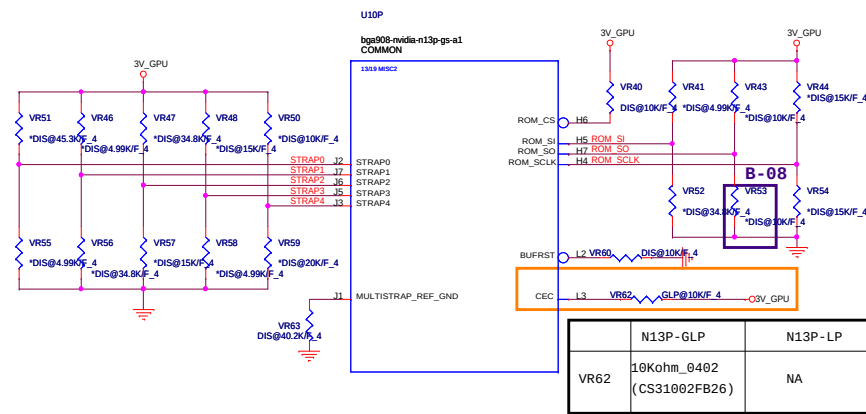
To be configured as needed on the PCB



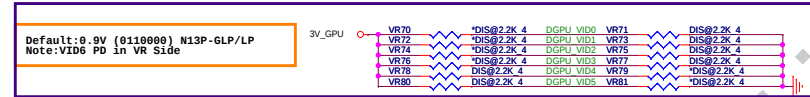
Quanta Computer Inc.
PROJECT : FH6C_HM70

Size	Document Number	Rev B
N13P GND		
Date	Tuesday, May 22, 2012	Sheet 20 of 45

1.Level 1 Environment-related Substances Should Never be Used.
2.Recycled Resin and Coated Wire should be procured from Green Partners.



C-33 For IVMP 6.5 (GPU)



Logical Strap Bit Mapping

Value	PU-VDD	PD	QCI PN(0402)
4.99K	1000	0000	CS24992FB26
10K	1001	0001	CS31002FB26
15K	1010	0010	CS31502FB24
20K	1011	0011	CS32002FB29
24.9K	1100	0100	CS32492FB16
30.1K	1101	0101	CS33012FB18
34.8K	1110	0110	CS33482FB22
45.3K	1111	0111	CS34532FB18

VRAM(DDR3) Configuration Table

RAMCFG [3:0]	DESCRIPTION (Vendor P/N)	Vendor	QCI P/N	ROM_SI
0111	128*16-900MHz K4W2G1646C-HC11	Samsung	AKD5MGWT500	PD 45.3K
0110	128*16-900MHz H5TQ2G63BFR-11C	Hynix	AKD5MGWTW00	PD 35K
0010	64*16-900MHz H5TQ1G63DFR-11C	Hynix	AKD5LZWTW02	PD 15K
0011	64*16-900MHz K4W1G1646G-BC11	Samsung	AKD5EGGT500	PD 20K

N13P-LP (GK107-ESP)	Logical Strapping Bit3	Logical Strapping Bit2	Logical Strapping Bit1	Logical Strapping Bit0
ROM_SCLK	PCI_DEVIDE[4]	SUB_VENDOR	SLOT_CLK_CFG-GLP	PEX_PLL_EN_TERM
ROM_SI	RAMCFG[3]	RAMCFG[2]	RAMCFG[1]	RAMCFG[0]
ROM_SO	XCLK_417 FB[1]-LP	FB_0_BAR_SIZE FB[0]-LP	SMB_ALT_ADDR I2CS_ADDR:0X9E	VGA_DEVICE
STRAP0	USER[3]	USER[2]	USER[1]	USER[0]
STRAP1	3GIO_PADCFG[3]	3GIO_PADCFG[2]	3GIO_PADCFG[1]	3GIO_PADCFG[0]
STRAP2	PCI_DEVID[3]	PCI_DEVID[2]	PCI_DEVID[1]	PCI_DEVID[0]
STRAP3	SOR3_EXPOSED	SOR2_EXPOSED	SOR1_EXPOSED	SOR0_EXPOSED
STRAP4	RESERVED	Reserve PCIE_SPEED-LP	PCIE_MAX_SPEED	DP_PLL_VDD3V

B-29

	GLP 1GB HYN	GLP 1GB SAM	GLP 2GB HYN	GLP 2GB SAM	LP 2GB HYN	LP 2GB SAM
ROM_SCLK	VR44 VR54 CS31502FB24	NA CS31502FB24	CS31502FB24	CS31502FB24	NA CS24992FB26	NA CS24992FB26
ROM_SI	VR41 VR52 CS31502FB24	NA CS32002FB29	CS33012FB18	CS34532FB18	NA CS33012FB18	NA CS34532FB18
ROM_SO	VR43 VR53 CS31002FB26	NA CS31002FB26	CS31002FB26	CS31002FB26	NA CS31002FB26	NA CS31002FB26
STRAP0	VR51 VR55 NA CS34532FB18	NA CS34532FB18	NA CS34532FB18	NA CS34532FB18	NA CS34532FB18	NA CS34532FB18
STRAP1	VR46 VR56 NA CS34532FB18	NA CS34532FB18	CS34532FB18	CS34532FB18	CS24992FB26	CS24992FB26
STRAP2	VR47 VR57 CS24992FB26	NA CS24992FB26	NA CS24992FB26	NA CS24992FB26	NA CS32002FB29	NA CS32002FB29
STRAP3	VR48 VR58 NA CS24992FB26	NA CS24992FB26	NA CS24992FB26	NA CS24992FB26	NA CS24992FB26	NA CS24992FB26
STRAP4	VR50 VR59 NA NA	NA NA	NA NA	NA NA	NA CS34532FB18	NA CS34532FB18

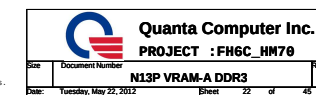
Value	Note
1000	LP:ES Samples 5K PU(0X0FDB)
0010	GLP:ES Samples 15K PD(0X0DFE)
0110	LP H5TQ2G63BFR-11C:35K PD
0111	K4W2G1646C-HC11:45.3K PD
0010	GLP H5TQ1G63DFR-11C:15K PD
0011	K4W1G1646G-BC11:20K PD
1001	10K PU
0001	10K PD
1111	EDID is used :45K PU
0000	LP:notebook default:35K PD
0111	GLP: Reserve:45.3K PD
0011	LP ES Samples:20K PU(0X0FDB)
1000	GLP ES Samples:45.3K PU(0X0DEF)
0000	Not in use :5K PD
0111	LP:10K PD
	GLP: NA

*LP QS Samples:5K PU(0X0FD3)
*GLP MP Samples 15K PD(0X0DE8)
*H5TQ2G63DFR-11C: 30.1K PD(0X05)
*LP QS Samples: 4.99K PD(0X0)
*LP QS Samples: 20K PD(0X0FD3)
*GLP MP Samples: 5K PU(0X0DE8)
*LP QS Samples: 45K PD(0X111)

C-48

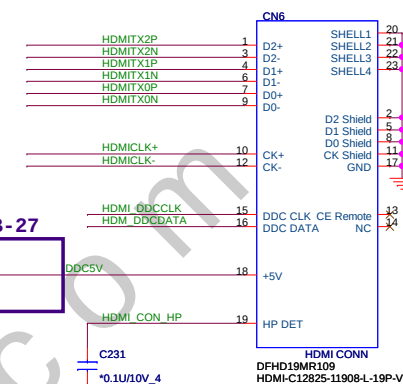
GPIO ASSIGNMENTS

GPIO	I/O	USAGE	
0	OUT	Y	NVDD VID4
1	OUT	Y	NVDD VID3
2	OUT	N	PANEL BACKLIGHT PWM
3	OUT	N	PANEL POWER ENABLE
4	OUT	N	PANEL BACKLIGHT ENABLE
5	OUT	Y	NVDD VID1
6	OUT	Y	NVDD VID2
7	OUT	N	3D STEREO
8	I/O	Y	GPU Overtemp
9	I/O	Y	GPU ALERT
10	OUT	N	FB Vref Control
11	OUT	Y	NVDD VID0
12	IN	N	PWR_Level AC Detect
13	OUT	Y	NVDD VID5
14	IN	N	HPD for IFP AB
15	IN	N	HPD for IFP C
16	OUT	N	DPRSLP(Default) or PSIF
17	OUT	N	HPD for IFP D
18	OUT	N	HPD for IFP E
19	OUT	N	HPD for IFP F
20	OUT	N	
21	OUT	N	





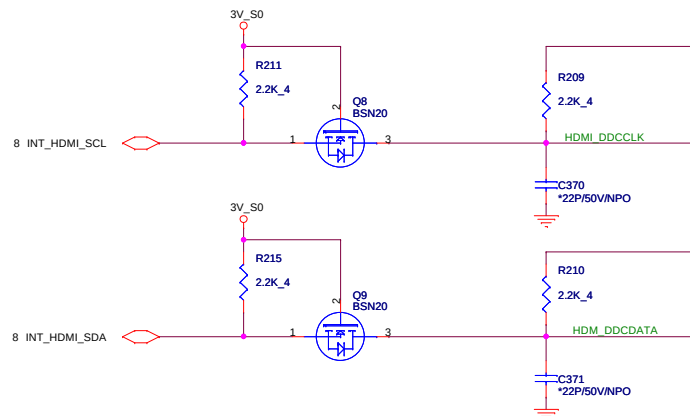
HDMI Conn



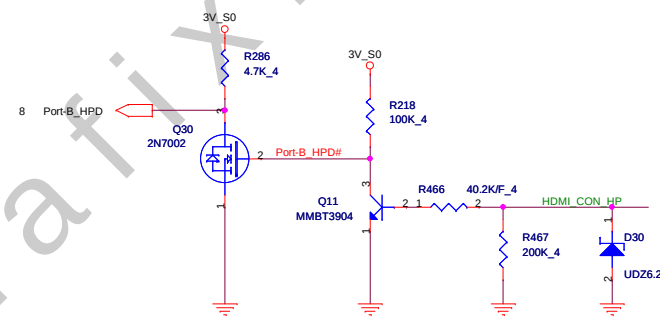
EMI



DDC Level Shift

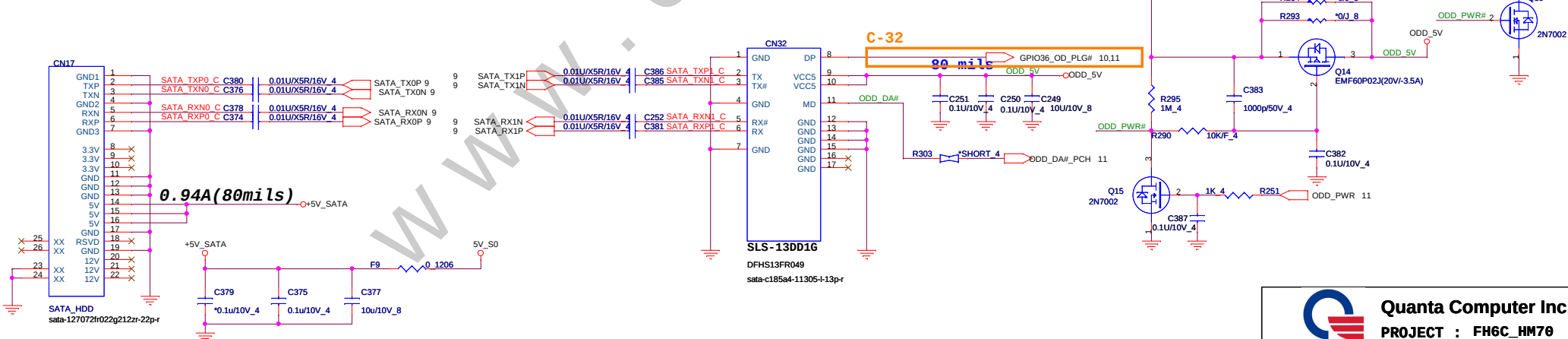


HPD



SATA ODD

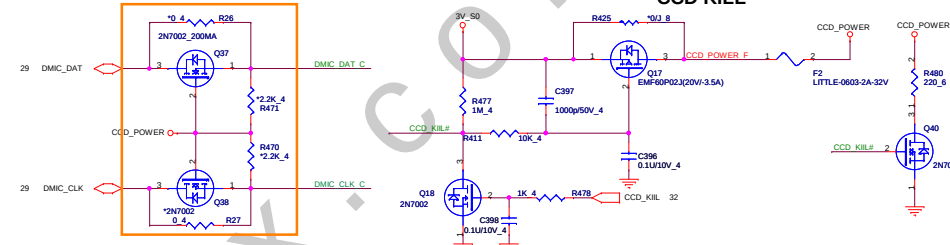
2.5" SATA HDD



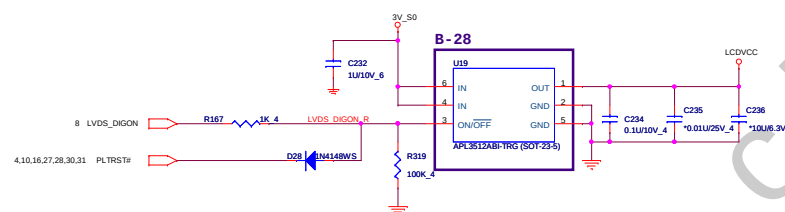
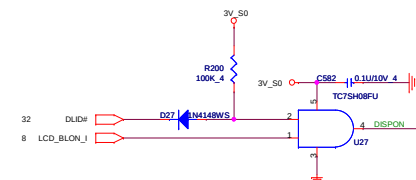
40mils



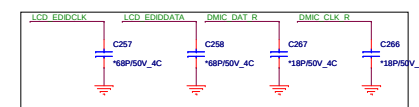
C-36



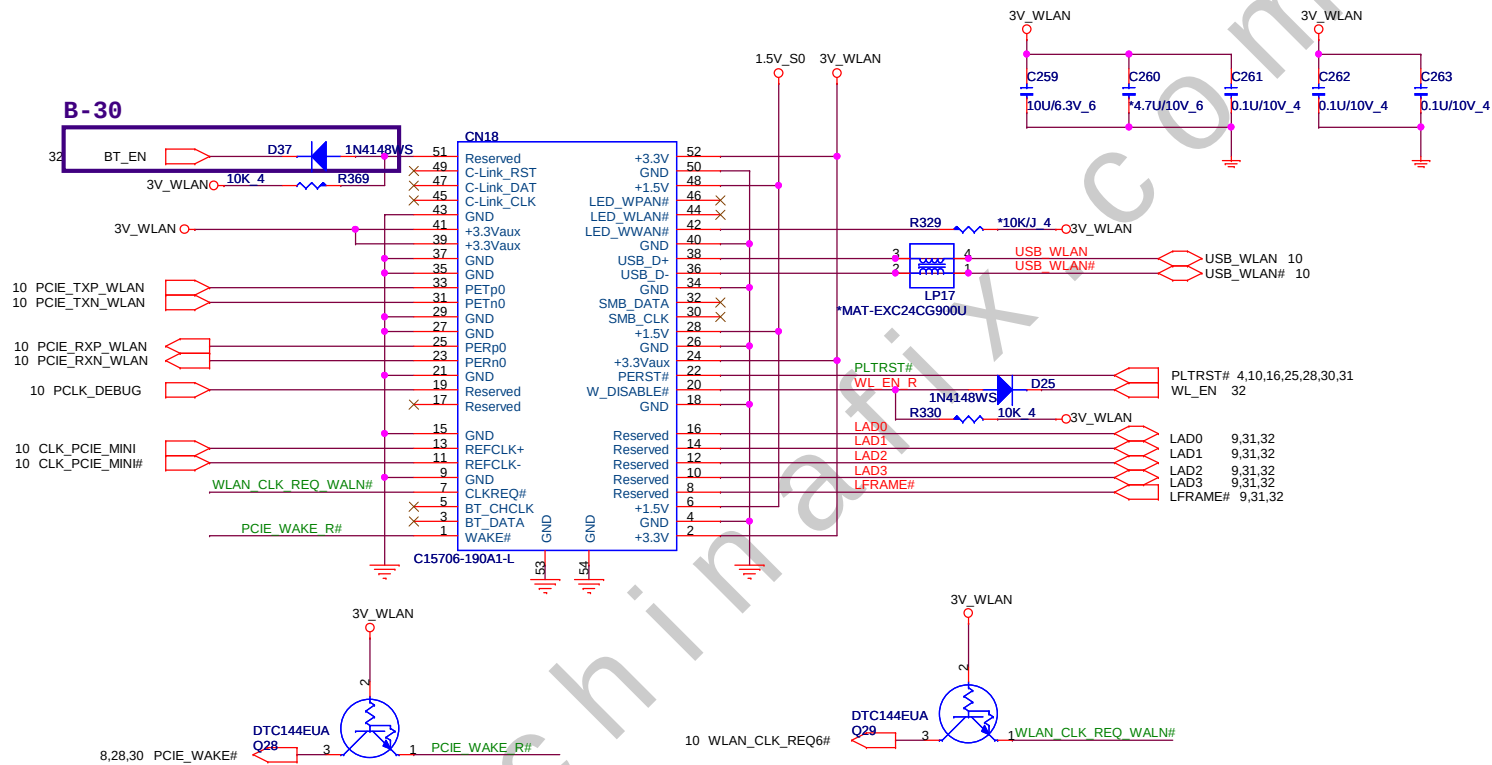
LCD Panel Module



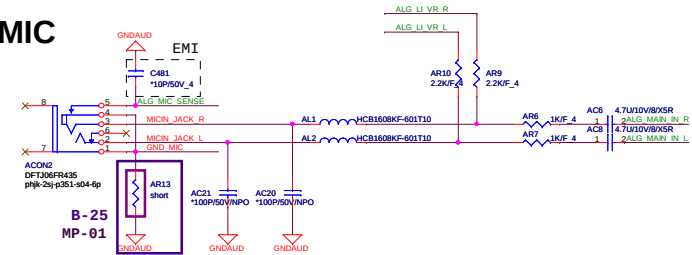
EMI



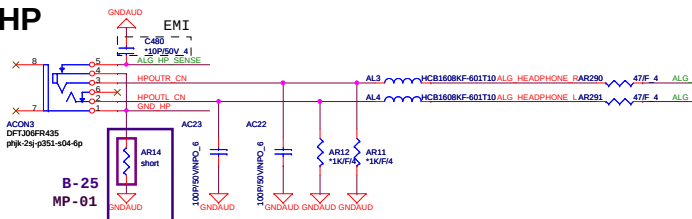
Wireless/BT COMBO



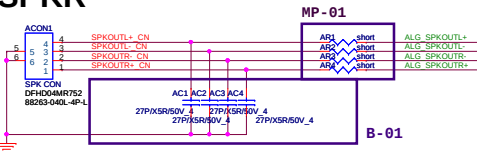
MIC



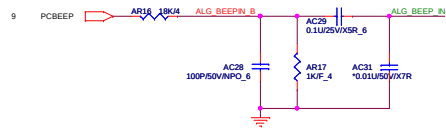
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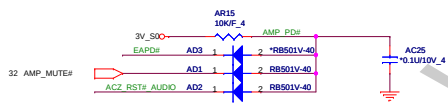
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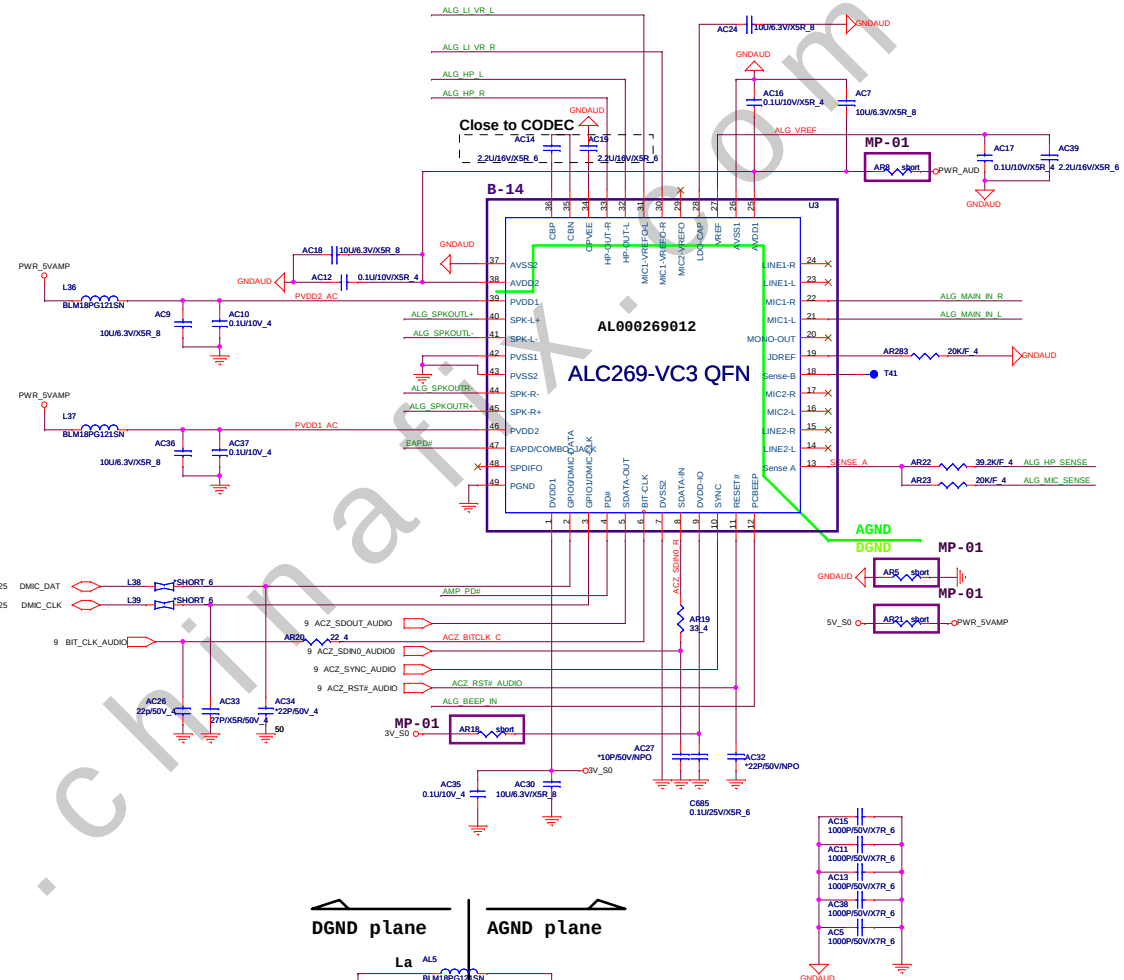
BEEP



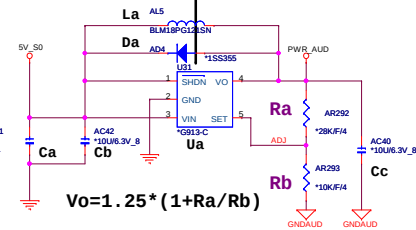
VOLMUTE

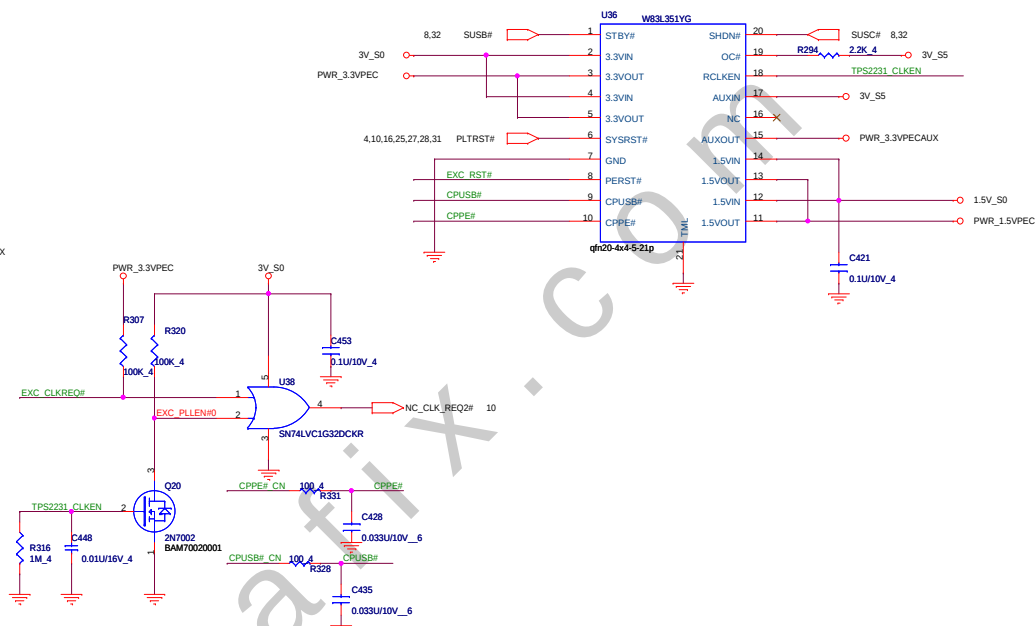


Codec ALC269-VC3



DGND plane AGND plane





10 USB_CARD#
10 USB_CARD

1 CHOKE WCM2012-900T 0.4A 100MHz

3V_S0

C644 *22P_4 U34

C641 *100P_NC 50

R475 5.2K 0.4

RREF

CARD_3V3

VREG

C613 4.7U6.3V_6 6.3 XSR 603

C610 0.1U/10V_4

C642 1U/10V_4

GND

RTSS138

TP82

TP88

TP80

250mA 30MIL

GARD_3V3 R465 SHORT_8 VCC_SD

VCC_SD

C393 10U/6.3V_8

C391 0.1U/10V_4

C388 0.1U/10V_4

C392 0.1U/10V_4

VCC_SD

SP14 MS_BS

SP12 MS_D1

SP9 MS_D0

SP8 MS_D2

SP7 MS_INS#

SP6 MS_D3

SP5 MS_CLK_C

SP1 R481 SHORT_4

C389 *22P_4

TP123

C394 *22P_4

R510 33.4

SD_CD# C

SD_CD# C

SD_SW(RSV) SD_SW(WP)

SD_SW(GND) SD_SW(WP-GND)

CN38

SD-CD/DAT3 SD-CMD

MS-BS MS-D1

MS-D0 MS-D2

MS-INS# MS-D3

MS-SCLK MS-CD# C

MS-VCC MS-VCC

MS-GND2

SD-D0

SD-D1

SD-D2

SD-D3 C

SD_CMD R

R515 33.4

R503 22.4

R516 33.4

R517 33.4

R518 33.4

R520 33.4

C390 *22P_4

SD D3 SP12

SD CMD R515 SP10

SD D0 SP8

SD D1 SP4

SD D2 SP3

SD D3 SP13

SD WP SP1

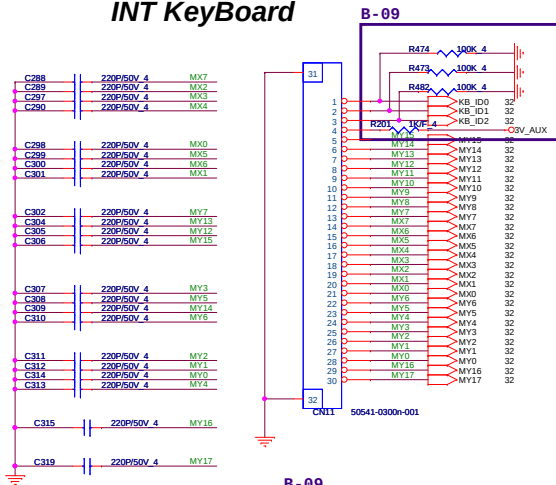
Supporting MMC/SD/MS Cards

SP1	SD_WP	MS_CLK
SP2		MS_INS#
SP3	SD_D1	
SP4	SD_D0	
SP5		MS_D3
SP6	SD_CD#	
SP8	SD_CLK	MS_D2
SP9		MS_D0
SP10	SD_CMD	
SP12	SD_D3	MS_D1
SP13		

SP1	SD WP	MS CLK
SP2		MS INS#
SP3	SD D1	
SP4	SD D0	
SP5		MS D3
SP6	SD CD#	
SP8	SD CLK	MS D2
SP9		MS D0
SP10	SD CMD	
SP12	SD D3	MS D1
SP13	SD D2	
SP14		MS BS

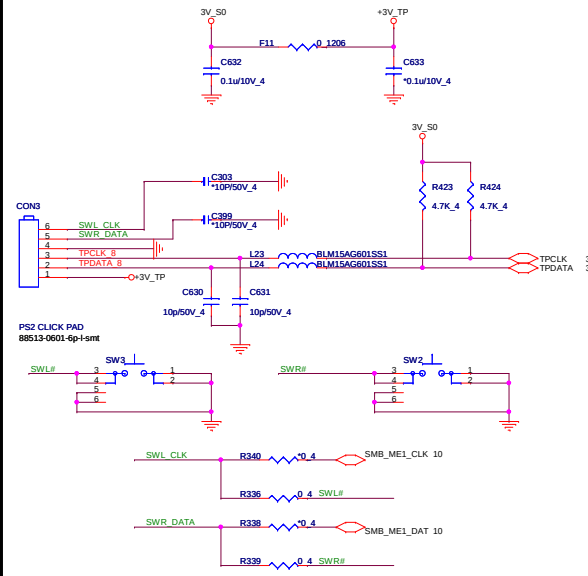
Share Pin

INT KeyBoard

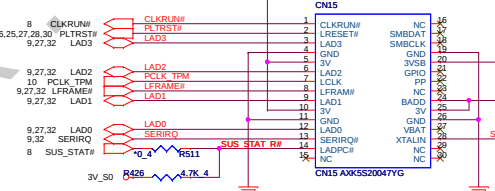


	ID0	ID1	ID2	KB_ID
	CHO	ISO		
UK	1	0	0	1
US	0	1	0	1
JP	1	1	0	1

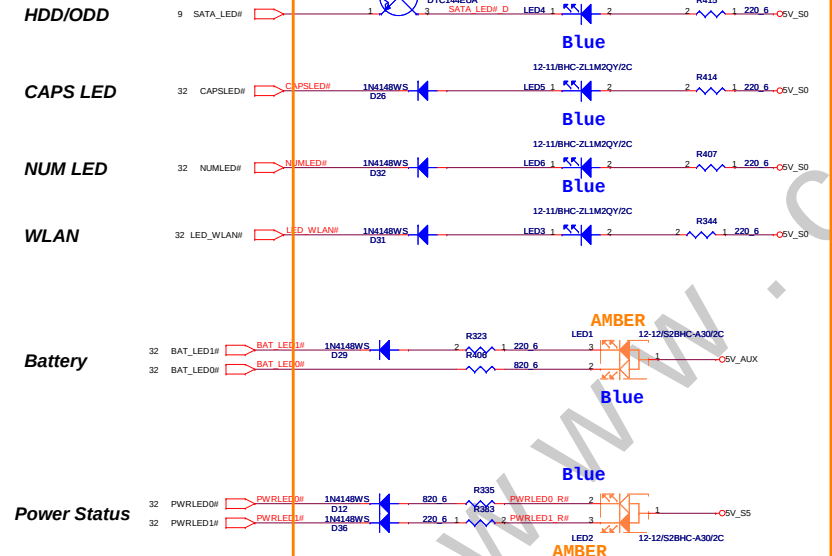
Touch Pad SMBUS CLICK PAD reserve



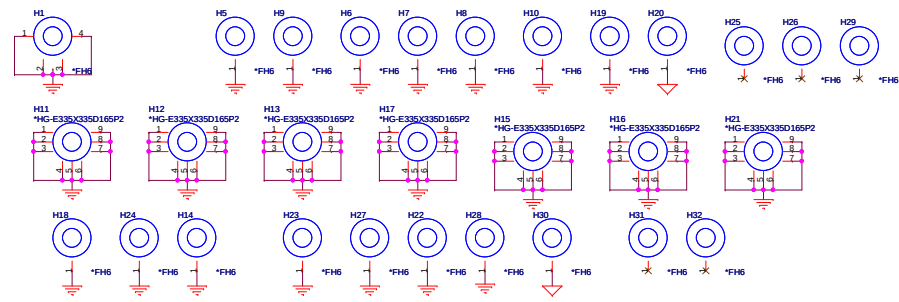
TPM Connector



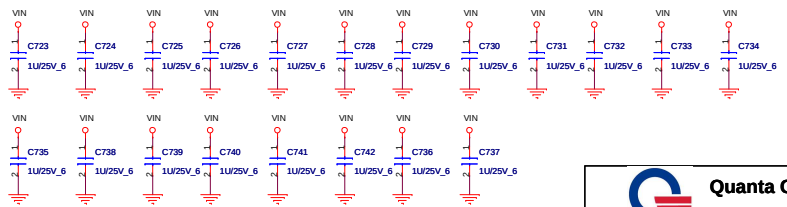
LED

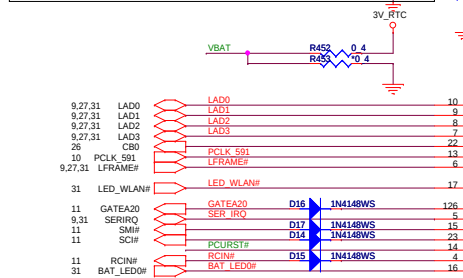
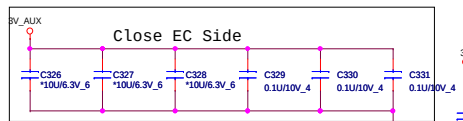


HOLE

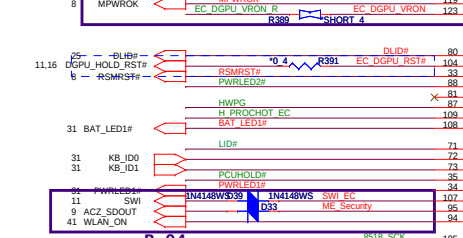


Decoupling Cap

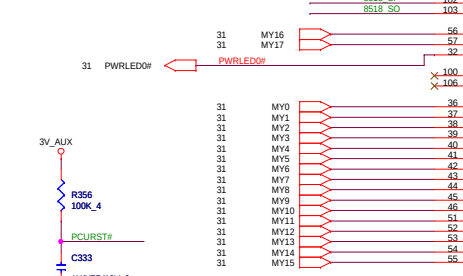




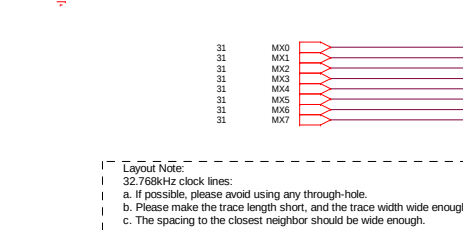
B-21



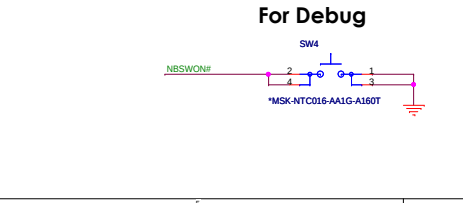
B-04



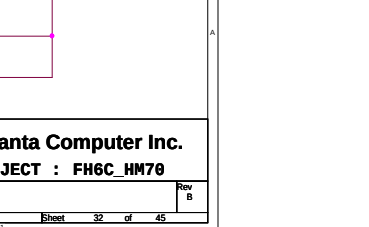
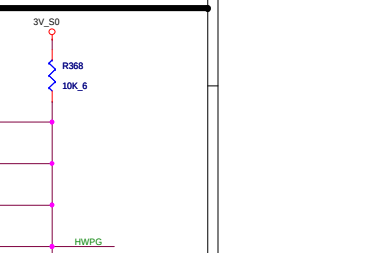
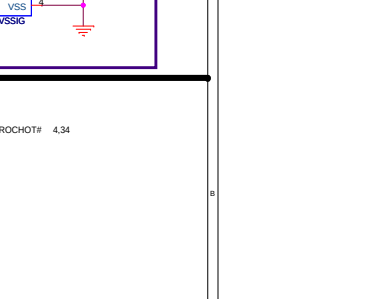
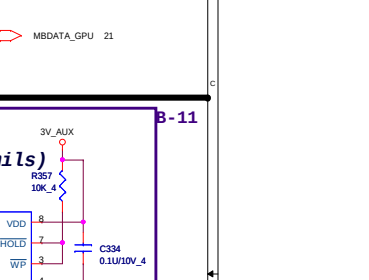
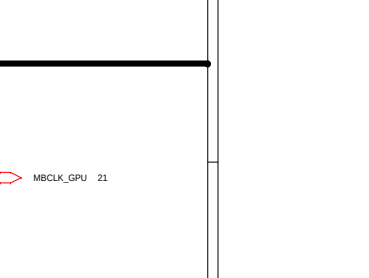
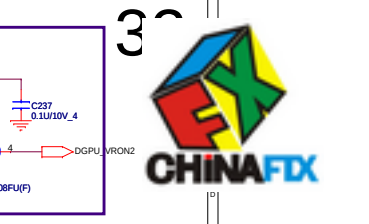
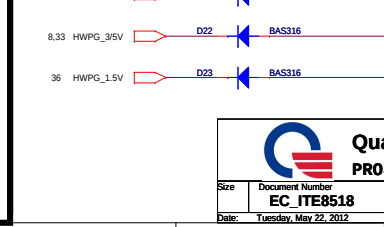
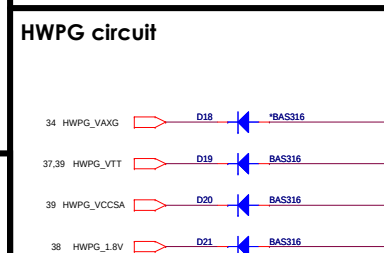
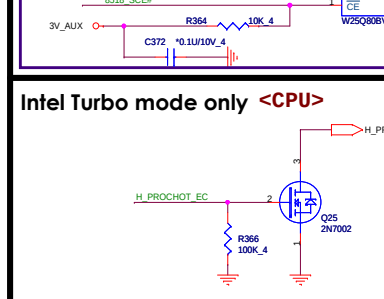
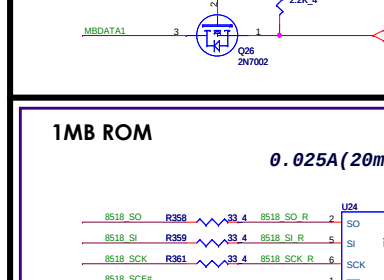
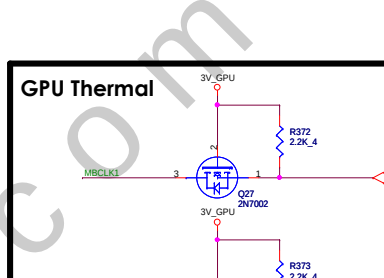
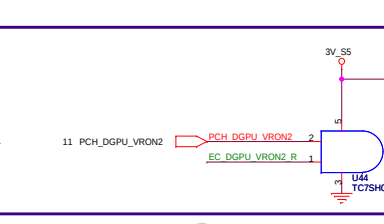
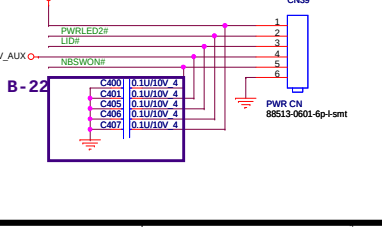
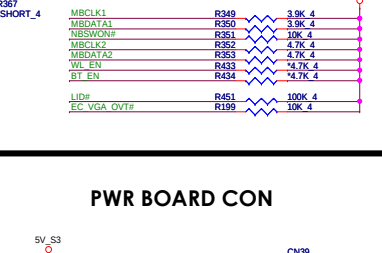
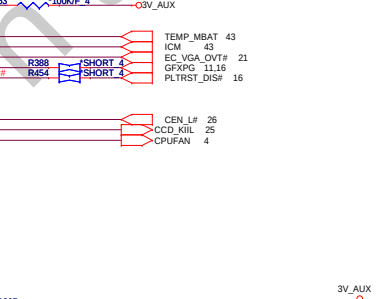
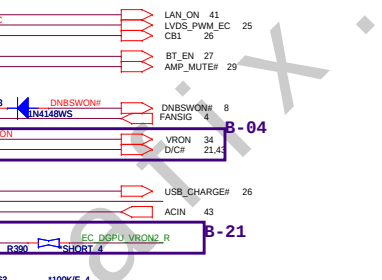
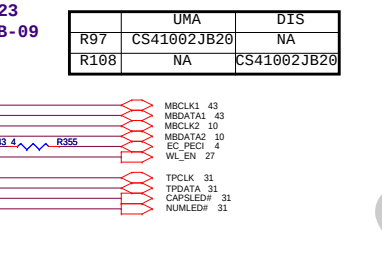
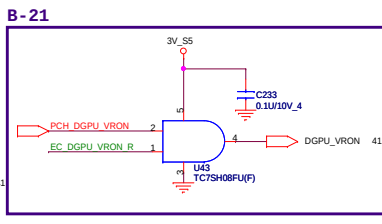
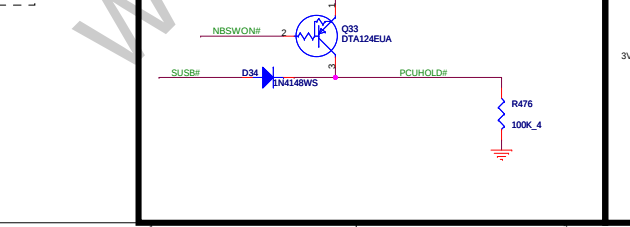
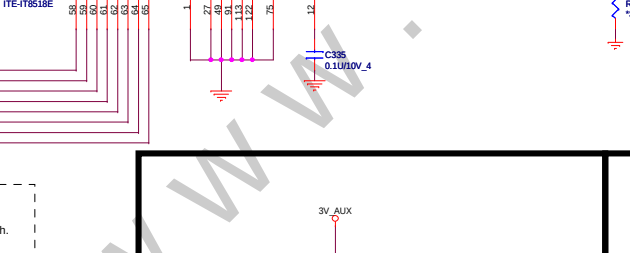
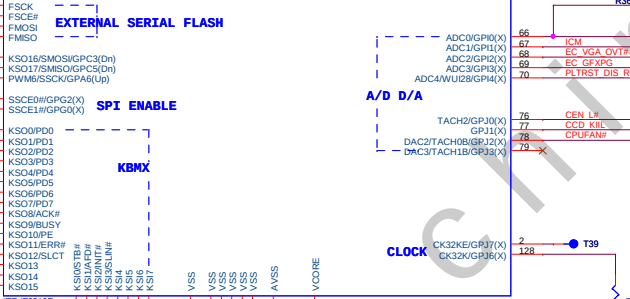
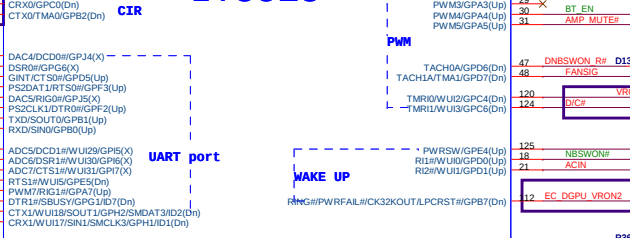
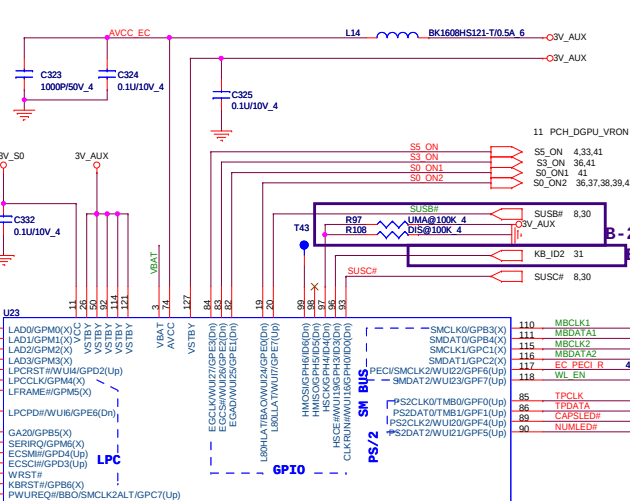
B-04

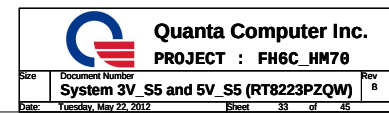


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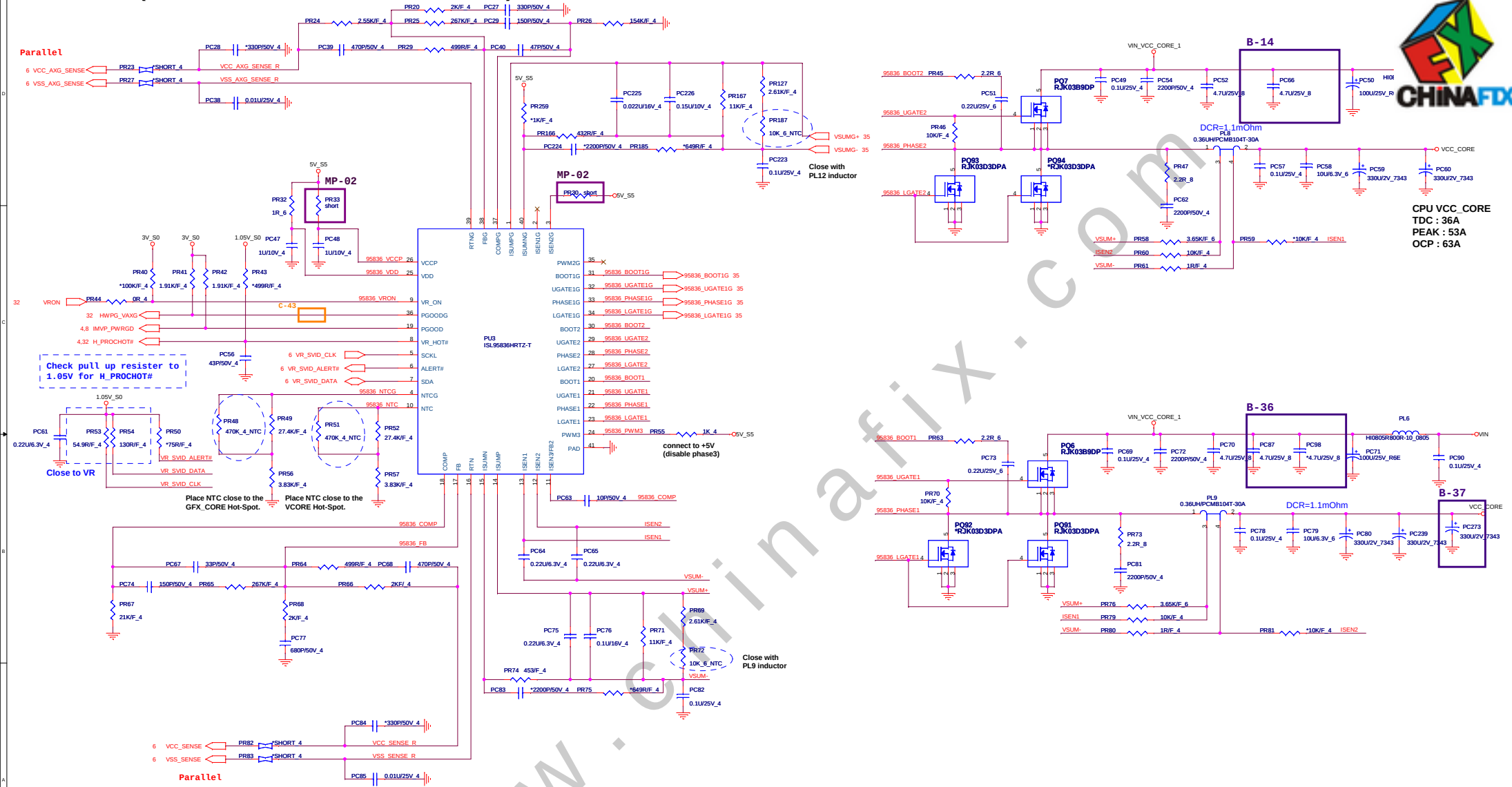


B-04





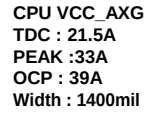
CPU VCORE (ISL95836HRTZ-T and ISL6208CRZ-T)



CPU VCC_CORE
TDC : 36A
PEAK : 53A
OCP : 63A

Inductor information

Value	Vendor	QCI P/N	Irms(A)	Isat(A)	Rdc (ohm)	Size	Vendor P/N
0.36uH 20%	CYN	CV+36V0MZ13	30	50	1.4m Max.	10x10x4	PCMB104T-R36MT
0.36uH 20%	Panasonic	CV+18V0MZ04	30	34	1.4m Max.	10x10x4	ETQP4LR36WFC



Value	Vendor	QCI P/N	Irms(A)	Isat(A)	Rdc (ohm)	Size	Vendor P/N
0.36uH 20%	Panasonic	CV+36Q0MZ00	20	25	1.4m Max.	7X7X4	ETQP4LR36AFM

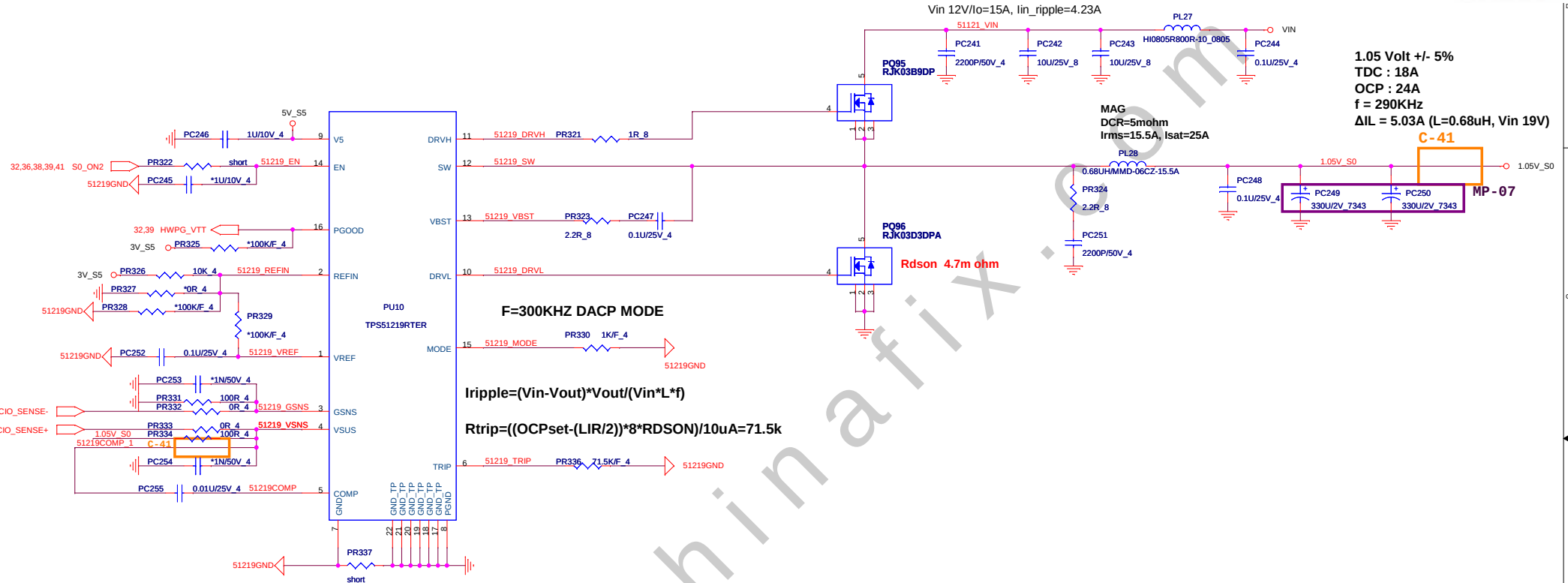
Mosfet	Package	ID (Ta=25C)	Rds_on_max	Schottky
RJK03D3DPA	P_PAK	20A/40A	4.7m	YES
AOL1718	P_PAK	20A/90A	4.3m	YES
RMW200N03FUB	P_PAK	20A/80A	4.6m	NO
FDMS0310S	P_PAK	14A/83A	5.2m	YES



Mosfet	Package	ID (Ta=25C)	Rds_on_max	Schottky
RJK03D3DPA	P_PAK	20A/40A	4.7m	YES
AOL1718	P_PAK	20A/90A	4.3m	YES
RMW200N03FUB	P_PAK	20A/80A	4.6m	NO
FDMS0310S	P_PAK	14A/83A	5.2m	YES

Value	Vendor	QCI P/N	Irms(A)	Isat(A)	Rdc (ohm)	Size
1uH 20%	CYN	CV-10I0MZ04	18	28	3.3m Max.	11X10X4
1uH 20%	MAG Layer	CV-10L0MZ28	21	30	3.1m Max.	11X10X4

VCCIO and 1.05V_PCH (TPS51219RTER)

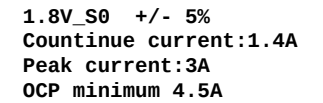


Output Voltage Selection	
RFIN=3.3V	output voltage=1.05V
RFIN=GND	output voltage=1.00V
Resister Divider	Adjustable from VREF

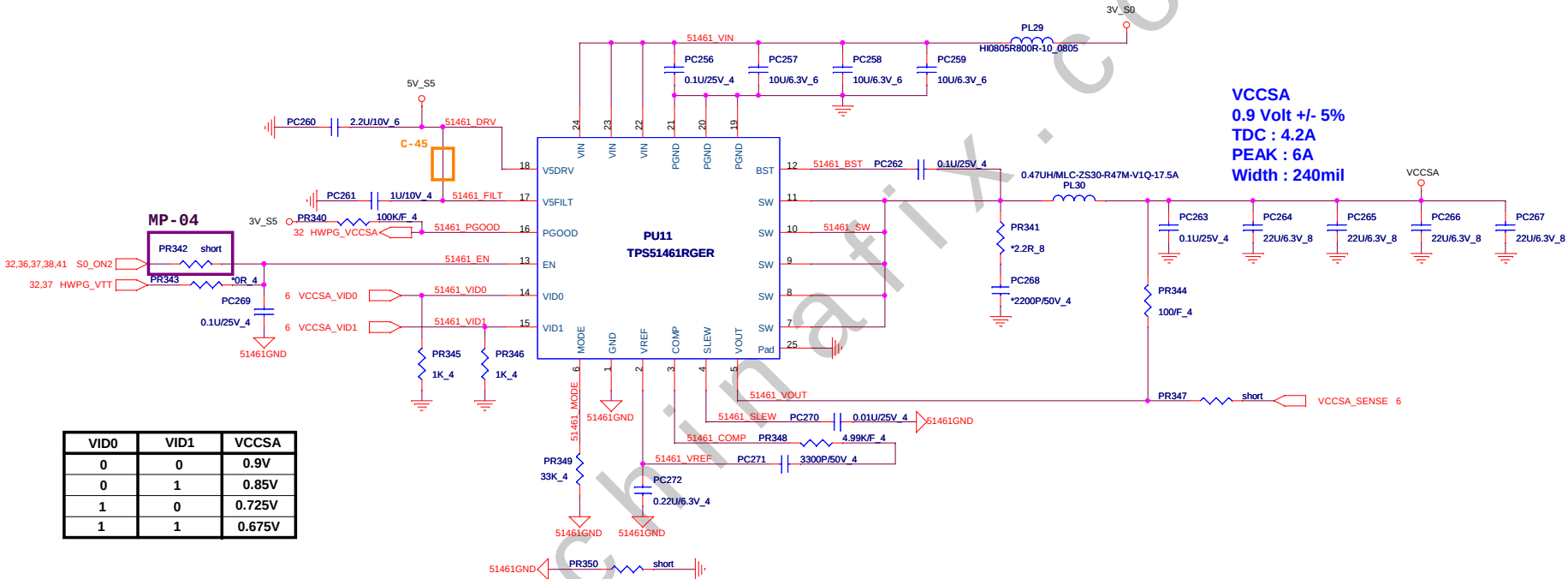
Inductor information					
Vendor	QCI P/N	Irms(A)	Isat(A)	Rdc (ohm)	Size
CYN	CV-10I0MZ04	18	28	3.3m Max.	11X10X4
MAG Layer	CV-10L0MZ28	21	30	3.1m Max.	11X10X4

O.C.P setup information						
Output	Mos Rds_on	I_OCP	OC_ΔIL(A)	Freq(KHz)	Inductor	R_TRIP
1.05V	4.3m_Max	24	3.306	300	1uH	56.2K

VCCIO_SENSE- connect to the GND sense point of the load
VCCIO_SENSE+ connect to the load voltage sense point.



VCCSA (TPS51461RGER)



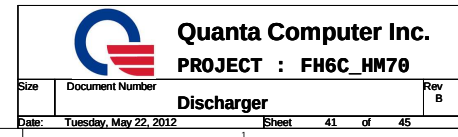
VCCSA
0.9 Volt +/- 5%
TDC : 4.2A
PEAK : 6A
Width : 240mil

VID0	VID1	VCCSA
0	0	0.9V
0	1	0.85V
1	0	0.725V
1	1	0.675V

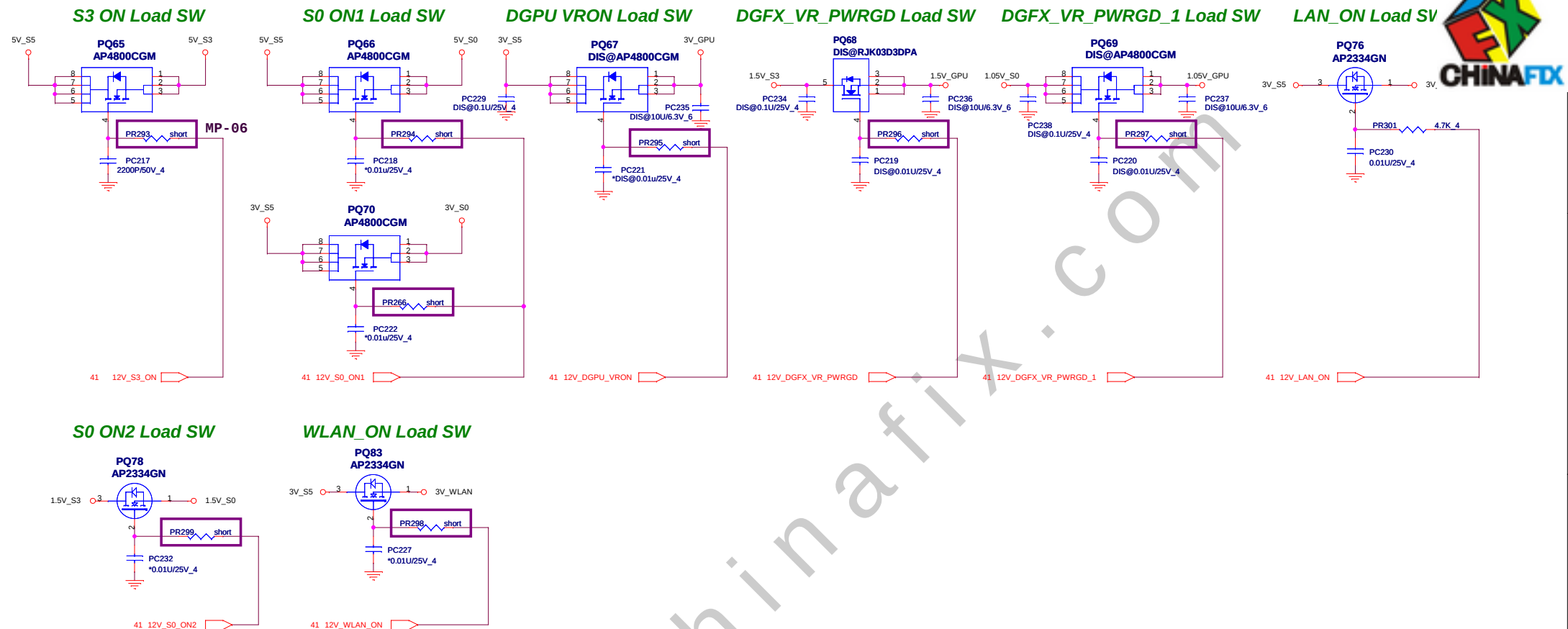


Quanta Computer Inc.
PROJECT : FH6C_HM70

Size	Document Number	Rev
	VCCSA (TPS51461RGER)	B



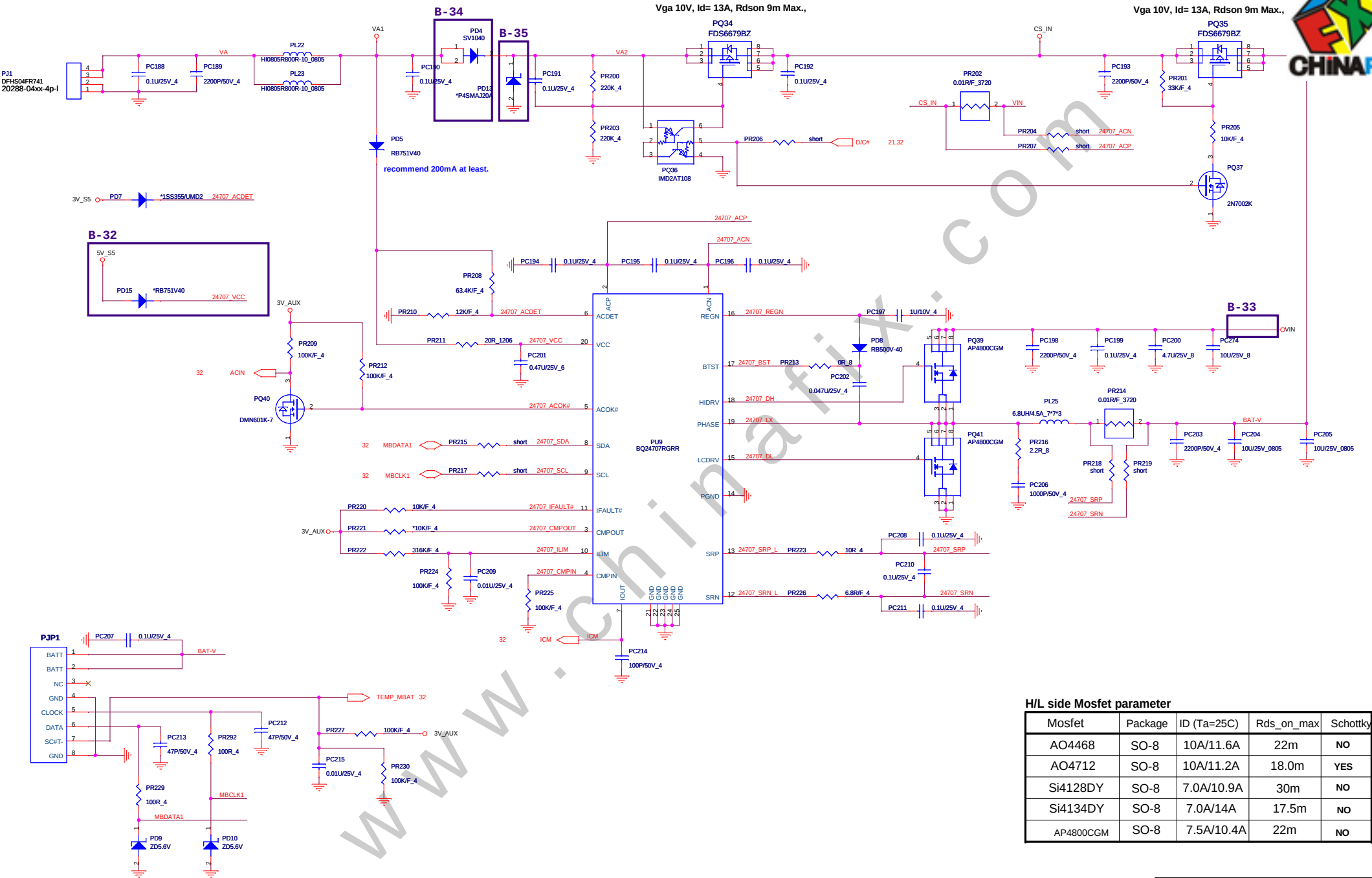
Load Switch



Mosfet parameter

Mosfet	Package	ID(Ta=25C)	Rds_on_max	Vgs_max
AO4468	SO-8	8.4A/10.4A	22m	+/- 20V
AP4800CGM	SO-8	7.5A/10.4A	22m	+/- 20V
Si4128DY	SO-8	7.0A/10.9A	30m	+/- 20V
Si4134DY	SO-8	7.0A/14A	17.5m	+/- 20V
ME3424D	TSOP-6	5.0A/6.7A	42m	+/- 20V
AP2334GN	SOT-23	4.5A/5.0A	42m	+/- 20V
AO3404	SOT-23	5.0A/5.8A	43m	+/- 20V

Adapter/Battery input & Charger



H/L side Mosfet parameter

Mosfet	Package	ID (Ta=25C)	Rds_on_max	Schottky
AO4468	SO-8	10A/11.6A	22m	NO
AO4712	SO-8	10A/11.2A	18.0m	YES
Si4128DY	SO-8	7.0A/10.9A	30m	NO
Si4134DY	SO-8	7.0A/14A	17.5m	NO
AP4800CGM	SO-8	7.5A/10.4A	22m	NO

[illegible]